



Ch 1: Chemical Reactions and Equations

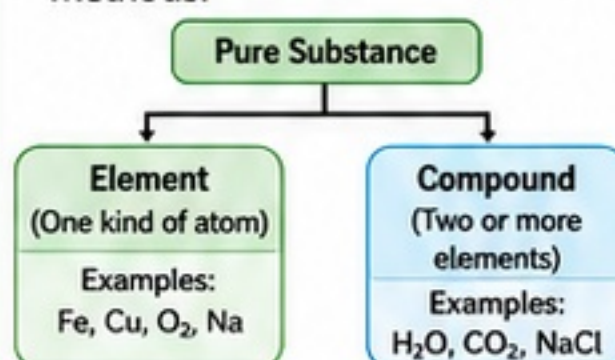
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SECTION A – INTRODUCTION TO CHEMICAL REACTIONS

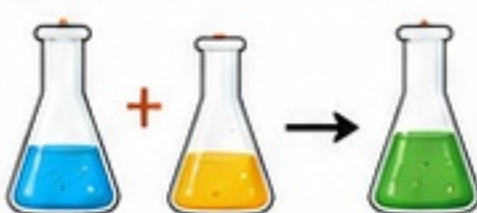
1 Matter and Substances (Very Brief Refresher)

- Matter:** Anything that has mass and occupies space.
- Pure substance:** Matter that has a fixed composition and distinct properties.
- Element:** A pure substance made of only one kind of atom. It cannot be broken down by chemical means.
- Compound:** A pure substance formed when two or more elements chemically combine in a fixed ratio. It can be decomposed by chemical methods.



2 Chemical Reaction

A chemical reaction is a process in which one or more substances (reactants) change into new substances (products) with different properties.



Meaning:

New substances are formed with different physical and chemical properties.

Example:

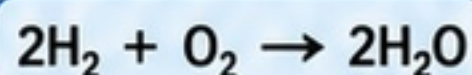
Hydrogen burns in oxygen to form water (steam).

3 Reactants and Products

Reactants: The starting substances that take part in a reaction.

Products: The new substances formed in a reaction.

Example:



Reactants

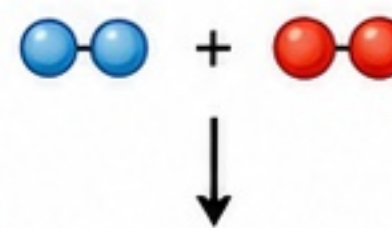
Product

Here, 2H_2 and O_2 are reactants. $2\text{H}_2\text{O}$ (water) is the product.

Water is a new substance with different properties from hydrogen and oxygen.

4 Rearrangement of Atoms

In a chemical reaction, atoms of the reactants are rearranged to form products.



● = Hydrogen (H)

● = Oxygen (O)

Atoms are neither created nor destroyed in a chemical reaction. They are only rearranged.

5 Law of Conservation of Mass

Mass can neither be created nor destroyed in a chemical reaction.

The total mass of reactants is always equal to the total mass of products.

Significance:

This law helps in balancing chemical equations and is fundamental to all chemical calculations.



6 Major Signs / Indicators of a Chemical Reaction

Sign / Indicator	What Happens	Example (with Colour / Observation)
Change in State	Change from one physical state to another (solid $\rightleftharpoons$ liquid $\rightleftharpoons$ gas).	Ice (solid) melts to water (liquid); water boils to steam (gas).
Change in Colour	One or more substances change their colour.	Copper sulphate solution is blue . When an iron nail is placed in it, the blue solution turns greenish (ferrous sulphate, FeSO_4) and reddish-brown copper (Cu) is deposited on the nail.
Change in Temperature	Heat is evolved (rise in temperature) or absorbed (fall in temperature).	When quicklime (CaO) is added to water, temperature rises (heat is evolved). When ammonium chloride dissolves in water, temperature falls (cold feeling).
Evolution of Gas	A gas is produced during the reaction (bubbles / effervescence seen).	Zinc reacts with dilute HCl to give hydrogen gas (colourless) with 'pop' sound on burning.
Formation of Precipitate	An insoluble solid (precipitate) separates out from the solution.	When silver nitrate solution is added to sodium chloride solution, white precipitate of silver chloride (AgCl) forms.
Change in Smell	A new smell is produced or an existing smell changes.	When ethanol burns, a pleasant smell is noticed. When food spoils, a foul smell develops.

7 Why Chemical Reactions Are Important in Daily Life

Combustion Burning of fuels (LPG, petrol, wood) to produce heat and energy.	Digestion Food is broken down by chemical reactions in our stomach.	Respiration Glucose reacts with oxygen in our cells to release energy.	Photosynthesis Plants use CO_2 and water to make glucose and release O_2 .	Rusting Iron reacts with oxygen and water to form rust (brown).	Medicines Many medicines work through chemical reactions in our body.	Industry Chemicals are manufactured to make products like plastics, soaps, fertilizers.	Cleaning Soaps and detergents remove dirt by chemical action.
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8 Quick Key Terms

Term	Definition
Matter	Anything that has mass and occupies space.
Pure Substance	Substance with fixed composition and distinct properties.
Element	Pure substance made of one kind of atom; cannot be broken down by chemical means. (e.g., Fe, Cu, O_2)
Compound	Pure substance of two or more elements in fixed ratio; can be decomposed chemically. (e.g., H_2O , CO_2 , NaCl)
Reactant	Starting substance that takes part in a chemical reaction.
Product	New substance formed in a chemical reaction.
Precipitate	Insoluble solid formed during a reaction in solution.

9 Exam Tips / Quick Recall

- ★ Remember: New substance + Different properties = Chemical Reaction.
- ★ In any reaction, atoms are only rearranged, not created or destroyed.
- ★ Total mass of reactants = Total mass of products (Law of Conservation of Mass).
- ★ Learn the signs of chemical reactions with examples and colours.
- ★ Practice writing and balancing simple equations.
- ★ Write colours of important compounds to score full marks in observations.
- ★ Diagrams (model dots/ball models, setups) help explain reactions better.
- ★ Always observe carefully in experiments—observation-based questions are common!





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SECTION B – CHEMICAL EQUATIONS

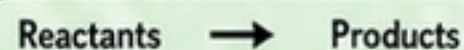
1 What is a Chemical Equation?

A chemical equation is a short way of writing a chemical reaction using symbols and formulae of substances.

Purpose:

- Shows what substances react and what substances are formed.
- Shows physical states and energy changes (where applicable).
- Helps in understanding, balancing and calculating reactions.

General form:



(Read as "React to form")

2 Types of Equations

(i) Word Equation:

Reactants and products are written in words.

(ii) Skeletal Equation:

Formulas of substances are written without balancing.

(iii) Balanced Equation:

Number of atoms of each element are equal on both sides (obeys law of conservation of mass).

3 Symbols that Build Equations

Symbol	Meaning
Na	Sodium (a metal, silvery-white in colour)
O	Oxygen (a non-metal, colourless gas)
Cl	Chlorine (a non-metal, greenish-yellow gas)
+	"Plus" sign: separates reactants or products
→	Arrow: "Yields" or "Forms"
(s)	Solid state
(l)	Liquid state
(g)	Gas state
(aq)	Aqueous state (dissolved in water)

4 Reactants and Products in Equation Form

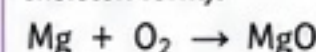
Reactants: written on the left side (LHS).

Products: written on the right side (RHS).

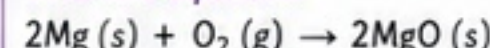
Example:

Magnesium + Oxygen → Magnesium oxide

In symbols (word equation to skeleton form):



Balanced equation:



Observation: Magnesium burns with a dazzling white flame and forms white ash (magnesium oxide).

5 Why Do We Need Chemical Equations?

- To represent reactions in a clear, universal and unambiguous form.
- To balance the number of atoms (law of conservation of mass).
- To help in calculations of amounts of reactants and products.
- To communicate scientific ideas across the world.

6 Law of Conservation of Mass and Equations



Mass can neither be created nor destroyed in a chemical reaction. Total mass of reactants = Total mass of products.

In Equations:

- Number of atoms of every element must be the same on LHS and RHS.
- This is achieved by using coefficients to balance equations.

Example: Iron + Sulphur → Iron sulphide
(iron sulphide is a black solid)

Wrong (not balanced): $\text{Fe} + \text{S} \rightarrow \text{FeS}$

Atoms count: Fe = 1 (LHS) ≠ 1 (RHS); S = 1 (LHS) ≠ 1 (RHS)

Balanced: $\text{Fe} (\text{s}) + \text{S} (\text{s}) \rightarrow \text{FeS} (\text{s})$

Atoms count: Fe = 1 = 1; S = 1 = 1 ✓

7 From Word Equation to Balanced Equation – Step by Step

Step	What to Do	Example: Combustion of Magnesium
1	Write the word equation.	Magnesium + Oxygen → Magnesium oxide (White ash is formed.)
2	Write the skeleton equation (using formulae).	$\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$
3	Balance metal first.	Already balanced for Mg (1 Mg on both sides).
4	Balance non-metal next.	O ₂ on LHS has 2 O atoms; MgO on RHS has 1 O. Put 2 before MgO. $\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
5	Check all elements (on both sides).	Mg = 1 = 2 (not balanced for Mg now) Correct by putting 2 before Mg. $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ Now Mg = 2 = 2 and O = 2 = 2 ✓
6	Add state symbols and check again.	$2\text{Mg} (\text{s}) + \text{O}_2 (\text{g}) \rightarrow 2\text{MgO} (\text{s})$ All atoms balanced. Equation is complete.

More Examples (Final Balanced Equations with Colour Info)

- Iron + Sulphur → Iron sulphide
 $\text{Fe} (\text{s}) + \text{S} (\text{s}) \rightarrow \text{FeS} (\text{s})$
(Black solid)
- Copper + Oxygen → Copper oxide
 $2\text{Cu} (\text{s}) + \text{O}_2 (\text{g}) \rightarrow 2\text{CuO} (\text{s})$
(Black solid)
- Hydrogen + Chlorine → Hydrogen chloride gas
 $\text{H}_2 (\text{g}) + \text{Cl}_2 (\text{g}) \rightarrow 2\text{HCl} (\text{g})$
(Colourless gas with pungent smell)
- Sodium + Chlorine → Sodium chloride
 $2\text{Na} (\text{s}) + \text{Cl}_2 (\text{g}) \rightarrow 2\text{NaCl} (\text{s})$
(White crystalline solid)
- Calcium oxide + Water → Calcium hydroxide
 $\text{CaO} (\text{s}) + \text{H}_2\text{O} (\text{l}) \rightarrow \text{Ca}(\text{OH})_2 (\text{aq})$
(CaO is white quicklime; Ca(OH)₂ forms a colourless solution)

8 Comparison of Word, Skeletal and Balanced Equations

Reaction	Word Equation	Skeletal Equation	Balanced Equation (with states & colour info)
1. Magnesium burns in oxygen	Magnesium + Oxygen → Magnesium oxide	$\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$	$2\text{Mg} (\text{s}) + \text{O}_2 (\text{g}) \rightarrow 2\text{MgO} (\text{s})$ (White ash of magnesium oxide)
2. Iron reacts with sulphur	Iron + Sulphur → Iron sulphide	$\text{Fe} + \text{S} \rightarrow \text{FeS}$	$\text{Fe} (\text{s}) + \text{S} (\text{s}) \rightarrow \text{FeS} (\text{s})$ (Black solid)
3. Copper reacts with dilute sulphuric acid	Copper + Dilute sulphuric acid → Copper sulphate + Hydrogen	$\text{Cu} + \text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{H}_2$	$\text{Cu} (\text{s}) + \text{H}_2\text{SO}_4 (\text{aq}) \rightarrow \text{CuSO}_4 (\text{aq}) + \text{H}_2 (\text{g})$ (Copper sulphate is a blue solution; Hydrogen is colourless gas that burns with a 'pop' sound)
4. Iron reacts with dilute sulphuric acid	Iron + Dilute sulphuric acid → Iron sulphate + Hydrogen	$\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2$	$\text{Fe} (\text{s}) + \text{H}_2\text{SO}_4 (\text{aq}) \rightarrow \text{FeSO}_4 (\text{aq}) + \text{H}_2 (\text{g})$ (Iron sulphate is a pale green solution; Hydrogen is colourless gas)
5. Silver nitrate reacts with sodium chloride	Silver nitrate + Sodium chloride → Silver chloride + Sodium nitrate	$\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$	$\text{AgNO}_3 (\text{aq}) + \text{NaCl} (\text{aq}) \rightarrow \text{AgCl} (\text{s}) + \text{NaNO}_3 (\text{aq})$ (Silver chloride is a white precipitate; Sodium nitrate is colourless solution)
6. Barium chloride reacts with sodium sulphate	Barium chloride + Sodium sulphate → Barium sulphate + Sodium chloride	$\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{NaCl}$	$\text{BaCl}_2 (\text{aq}) + \text{Na}_2\text{SO}_4 (\text{aq}) \rightarrow \text{BaSO}_4 (\text{s}) + 2\text{NaCl} (\text{aq})$ (Barium sulphate is a white precipitate; Sodium chloride is colourless solution)
7. Calcium carbonate on heating	Calcium carbonate → Calcium oxide + Carbon dioxide	$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$	$\text{CaCO}_3 (\text{s}) \rightarrow \text{CaO} (\text{s}) + \text{CO}_2 (\text{g})$ (Calcium oxide is white quicklime; Carbon dioxide is colourless gas)
8. Hydrogen burns in oxygen	Hydrogen + Oxygen → Water	$\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$	$2\text{H}_2 (\text{g}) + \text{O}_2 (\text{g}) \rightarrow 2\text{H}_2\text{O} (\text{l})$ (Water is colourless liquid; steam is colourless gas)

9 Memory Tip

"Left Reacts → Right Forms"
"Count Atoms – Match Atoms"
"Balance with Coefficients, not with subscripts!"



10 Common Mistakes

- ✗ Changing subscripts instead of using coefficients.
- ✗ Not balancing oxygen or hydrogen.
- ✗ Forgetting state symbols.
- ✗ Not checking all elements.
- ✗ Writing charges incorrectly in ionic reactions.

11 Exam Tips

- ★ Always write the balanced equation with states.
- ★ Mention colours/observations wherever asked.
- ★ Follow the step-by-step method.
- ★ Use coefficients in the lowest whole number ratio.
- ★ Practice common reactions like combination, decomposition, single displacement, double displacement.

12 Quick Recall

- ✓ LHS = Reactants, RHS = Products
- ✓ Total atoms (each element) must be equal on both sides.
- ✓ Use formulae, symbols and states.
- ✓ Colours/observations give strong hints in exam answers.



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SECTION C – BALANCING CHEMICAL EQUATIONS, STATES AND OBSERVATIONS

1 Why Balancing is Necessary

- Follows the Law of Conservation of Mass.
- Equal number of atoms of each element on both sides.
- Helps in correct calculations and understanding of reactions.



2 Rules for Balancing Equations

- Write correct and complete formulas.
- Count the number of atoms of each element on both sides.
- Balance one element at a time.
- Never change subscripts, only coefficients.
- Reduce coefficients to the simplest whole number ratio.



3 State Symbols

(s)	Solid Fixed shape and volume.
(l)	Liquid Fixed volume, no fixed shape.
(g)	Gas No fixed shape or volume.
(aq)	Aqueous Dissolved in water.

Example: $\text{CuSO}_4(\text{aq})$, $\text{NaCl}(\text{s})$, $\text{H}_2(\text{g})$, $\text{H}_2\text{O}(\text{l})$

4 Special Symbols

Δ	Heat
	Light
cat.	Catalyst
$\uparrow$	Gas evolved
$\downarrow$	Precipitate formed

Example:



5 Colours of Important Compounds / Products

CuSO_4 – blue
 FeSO_4 – pale green
 Cu – reddish-brown
 AgCl – white
 PbI_2 – yellow
 CuO – black
 MgO – white
 BaSO_4 – white
 Cl_2 – greenish-yellow gas
 H_2 – colourless gas
 CO_2 – colourless gas

6 Example 1: Balancing Step by Step

Balance the equation: $\text{Cu} + \text{O}_2 \rightarrow \text{CuO}$

Step	What We Do	Equation After Step
1	Count atoms: Cu = 1, O = 2 on left; Cu = 1, O = 1 on right.	$\text{Cu} + \text{O}_2 \rightarrow \text{CuO}$
2	Balance oxygen by putting 2 before CuO.	$\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$
3	Now Cu atoms = 2 on right. Put 2 before Cu.	$2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$
4	Recheck: Cu = 2 on both sides. O = 2 on both sides.	$2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$
5	Balanced!	$2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$

Balanced Equation: $2\text{Cu}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{CuO}(\text{s})$

Colours: Cu – reddish-brown, O_2 – colourless gas, CuO – black

7 Example 2: Balancing Step by Step

Balance the equation: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$

Step	What We Do	Equation After Step
1	Count atoms: Fe = 1, Cu = 1, S = 1, O = 4 on both sides.	$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$
2	All atoms are already equal. No change needed.	$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$
3	Check again: Fe = 1, Cu = 1, S = 1, O = 4 on both sides.	$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$
4	Already balanced.	$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$
5	Balanced!	$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$

Balanced Equation: $\text{Fe}(\text{s}) + \text{CuSO}_4(\text{aq}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$

Colours: CuSO_4 – blue, FeSO_4 – pale green, Cu – reddish-brown

8 Observation Table (During Reactions)

Reaction	Observation / Colour Change
$\text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$	Blue solution (CuSO_4) turns pale green (FeSO_4). Reddish-brown Cu deposits on iron.
$\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$	White precipitate (AgCl) is formed.
$\text{Pb}(\text{NO}_3)_2(\text{aq}) + 2\text{KI}(\text{aq}) \rightarrow \text{PbI}_2(\text{s}) + 2\text{KNO}_3(\text{aq})$	Yellow precipitate (PbI_2) is formed.
$2\text{HCl}(\text{aq}) + \text{MnO}_2(\text{s}) \rightarrow \text{MnCl}_2(\text{aq}) + \text{Cl}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	Greenish-yellow gas (Cl_2) is evolved.
$\text{Zn}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{ZnCl}_2(\text{aq}) + \text{H}_2(\text{g})$	Colourless gas (H_2) is evolved.
$\text{Na}_2\text{CO}_3(\text{aq}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$	Colourless gas (CO_2) is evolved; turns lime water milky.
$\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$	White precipitate (BaSO_4) is formed.

9 Commonly Asked Observation Examples

- ★ Blue CuSO_4 solution + Fe $\rightarrow$ pale green FeSO_4 solution and reddish-brown Cu forms.
- ★ White AgCl precipitate turns grey on exposure to sunlight.
- ★ Yellow PbI_2 dissolves in hot water and reappears on cooling.
- ★ Cl_2 gas bleaches moist litmus paper.
- ★ CO_2 gas turns lime water milky.
- ★ CuO is black; MgO and BaSO_4 are white.

Colour Key (Quick Look)

CuSO_4 – blue
 FeSO_4 – pale green
 Cu – reddish-brown
 AgCl – white
 PbI_2 – yellow
 CuO – black
 MgO – white
 BaSO_4 – white
 Cl_2 – greenish-yellow gas
 H_2 – colourless gas
 CO_2 – colourless gas

10 Common Mistakes

- ✗ Changing subscripts instead of coefficients.
- ✗ Not balancing one element at a time.
- ✗ Forgetting to balance polyatomic ions as a unit.
- ✗ Not checking the final count of atoms.
- ✗ Ignoring state symbols.
- ✗ Writing incomplete or incorrect formulas.
- ✗ Forgetting to reduce coefficients to the simplest ratio.

11 Exam Tips

- ★ Always write state symbols.
- ★ Balance equations neatly and systematically.
- ★ Learn common observations and colours.
- ★ Remember special symbols: Δ , light, cat., $\uparrow$, $\downarrow$.
- ★ Write balanced equations in answers for full marks.
- ★ Diagrams/colour labels can fetch extra marks.

12 Quick Check – Are You Ready?

1	I can balance simple chemical equations correctly.	☒
2	I know and use state symbols (s, l, g, aq).	☐
3	I can identify special symbols and their meanings.	☐
4	I can explain observations and colour changes in reactions.	☐
5	I remember the colours of important compounds and products.	☐
6	I can avoid common mistakes in equations.	☐

13 Centum Focus – High Marks Strategy

- ✓ Practise balancing at least 20–25 equations.
- ✓ Memorise colours and observations with examples.
- ✓ Draw small diagrams if needed for clarity.
- ✓ Write neat, well-balanced equations with correct symbols.
- ✓ Revise Tables 8, 9 and Colour Key daily.
- ✓ Solve NCERT in-text questions and exercise problems.





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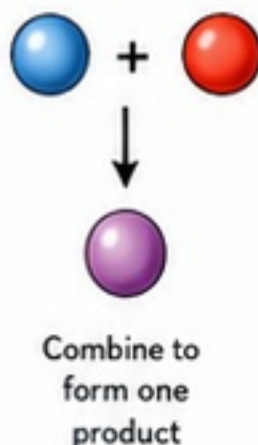


SECTION D – TYPES OF REACTIONS I: COMBINATION AND DECOMPOSITION

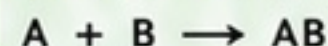
1 Combination Reaction

Two or more substances combine to form a single product.

- Usually exothermic.
- A new substance with different properties is formed.



General Form



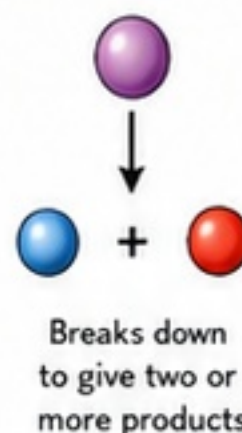
Key Features

- Increase in mass of the product.
- Heat or light may be evolved.
- The product is one substance.

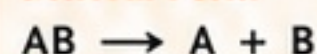
2 Decomposition Reaction

A single compound breaks down into two or more simpler substances.

- Usually endothermic.
- Requires energy (heat, electricity or light).



General Form



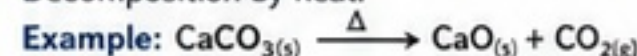
Key Features

- Decrease in mass of the compound.
- Two or more products are formed.
- Products may be elements or simpler compounds.

3 Types of Decomposition Reactions

(i) Thermal Decomposition

Decomposition by heat.



White calcium carbonate $\rightarrow$ white calcium oxide (quicklime) + colourless CO_2 gas.

(ii) Electrolytic Decomposition

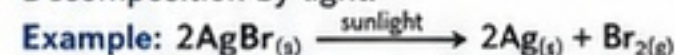
Decomposition by passing electric current.



Water decomposes to colourless hydrogen and colourless oxygen gases.

(iii) Photolytic Decomposition

Decomposition by light.



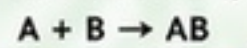
Silver bromide (pale yellow/cream) $\rightarrow$ silver (grey) + brown bromine vapour.

4 Classification Flow Diagram

Chemical Reactions

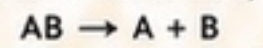
Combination

(Two or more $\rightarrow$ One)

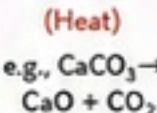


Decomposition

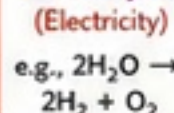
(One $\rightarrow$ Two or more)



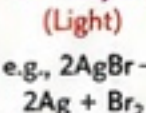
Thermal (Heat)



Electrolytic (Electricity)



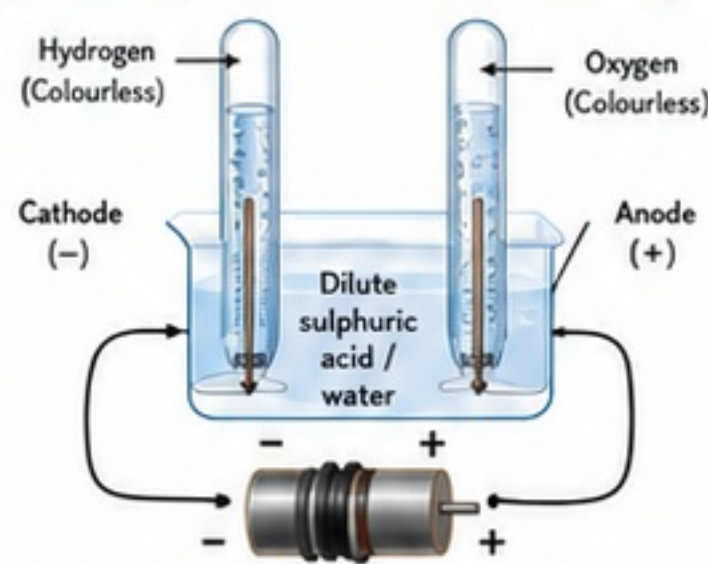
Photolytic (Light)



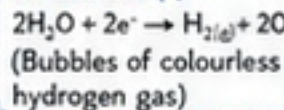
5 Examples Table

Type of Reaction	Balanced Equation	Products	Conditions	Observations	Colours of Compounds / Products
A. Combination					
1. Magnesium + Oxygen	$2\text{Mg}_{(s)} + \text{O}_{2(g)} \rightarrow 2\text{MgO}_{(s)}$	Magnesium oxide	Burning magnesium ribbon in air	Brilliant white light; white ash formed	White magnesium oxide ash
2. Calcium + Oxygen	$2\text{Ca}_{(s)} + \text{O}_{2(g)} \rightarrow 2\text{CaO}_{(s)}$	Calcium oxide	Burning calcium ribbon	Intense white light; white solid formed	White calcium oxide (quicklime)
3. Hydrogen + Chlorine	$\text{H}_{2(g)} + \text{Cl}_{2(g)} \rightarrow 2\text{HCl}_{(g)}$	Hydrogen chloride	Direct combination in sunlight	Dense white fumes	Colourless H_2 , greenish-yellow $\text{Cl}_2 \rightarrow$ colourless HCl gas
B. Decomposition – Thermal					
4. Calcium Carbonate	$\text{CaCO}_{3(s)} \xrightarrow{\Delta} \text{CaO}_{(s)} + \text{CO}_{2(g)}$	Calcium oxide + Carbon dioxide	Strong heating	Effervescence (CO_2 evolved); mass decreases	White $\text{CaCO}_3 \rightarrow$ white CaO + colourless CO_2 gas
B. Decomposition – Electrolytic					
5. Electrolysis of Water	$2\text{H}_2\text{O}_{(l)} \xrightarrow{\text{electricity}} 2\text{H}_{2(g)} + \text{O}_{2(g)}$	Hydrogen + Oxygen	Electric current	Bubbles at both electrodes	Colourless H_2 and colourless O_2 gases
B. Decomposition – Photolytic					
6. Silver Bromide	$2\text{AgBr}_{(s)} \xrightarrow{\text{sunlight}} 2\text{Ag}_{(s)} + \text{Br}_{2(g)}$	Silver + Bromine	Sunlight	Brown vapour evolved	Silver bromide pale yellow/cream; brown bromine vapour
7. Mercuric Oxide	$2\text{HgO}_{(s)} \xrightarrow{\Delta} 2\text{Hg}_{(l)} + \text{O}_{2(g)}$	Mercury + Oxygen	Heating	Silver droplets seen	Mercury is silver liquid; colourless O_2 gas
8. Copper(II) Carbonate	$\text{CuCO}_{3(s)} \xrightarrow{\Delta} \text{CuO}_{(s)} + \text{CO}_{2(g)}$	Copper(II) oxide + Carbon dioxide	Heating	Gas evolved; black solid remains	Black copper oxide; colourless CO_2 gas

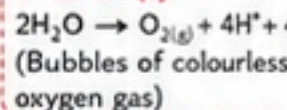
6 Electrolysis of Water – Detailed Diagram



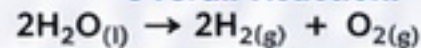
At Cathode (-):



At Anode (+):



Overall Reaction:



7 Comparison – Combination vs Decomposition

Feature	Combination Reaction	Decomposition Reaction
Definition	Two or more substances combine to form one product.	One compound breaks down to form two or more products.
General Form	$A + B \rightarrow AB$	$AB \rightarrow A + B$
Number of Reactants	Two or more	One
Number of Products	One	Two or more
Energy Change	Usually releases energy (heat or light evolved)	Usually requires energy (heat, electricity or light)
Change in Mass	Mass increases	Mass decreases
Examples	$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ $\text{Ca} + \text{O}_2 \rightarrow \text{CaO}$	$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

9 Exam Tips

- Write balanced chemical equations with states.
- Always mention conditions (heat Δ , electricity, sunlight).
- Write colours and observations – high-scoring points.
- For decomposition, identify the type (thermal, electrolytic, photolytic) first.
- Use arrows and symbols correctly ($\rightarrow$, Δ , $\downarrow$, $\uparrow$, state symbols).
- Practise common reactions: CaCO_3 , $2\text{H}_2\text{O}$, 2AgBr , HgO , CuCO_3 .

10 Common Mistakes

- Writing wrong formulas (e.g., $\text{CaO} \rightarrow \text{Ca}_2\text{O}$).
- Not balancing equations.
- Forgetting states (s, l, g, aq).
- Missing conditions (heat, electricity, sunlight).
- Not mentioning colours of products.
- Confusing combination with decomposition.
- Not showing precipitate ($\downarrow$) or gas ($\uparrow$) symbols.

11 Quick Recall

- ✓ **Combination:** Two or more $\rightarrow$ One ($A + B \rightarrow AB$)
- ✓ **Decomposition:** One $\rightarrow$ Two or more ($AB \rightarrow A + B$)
- ✓ **Types of Decomposition:** Thermal, Electrolytic, Photolytic
- ✓ White MgO ash, White CaO (quicklime), Black CuO
- ✓ Colourless H_2 and O_2 gases
- ✓ Brown Br_2 vapour, Greenish-yellow Cl_2 gas
- ✓ Silver bromide – pale yellow/cream; Silver chloride – white
- ✓ Mercury – silver liquid droplets

Colours of Important Compounds / Substances

- White MgO (magnesium oxide ash)
- White CaO (calcium oxide)
- Colourless H_2 gas
- Colourless O_2 gas
- Greenish-yellow Cl_2 gas
- Brown Br_2 vapour
- Silvery bromide – pale yellow/cream
- Silver chloride – white
- Mercury – silver liquid droplets
- Black CuO (copper(II) oxide)



Ch 1: Chemical Reactions and Equations

Part 5 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

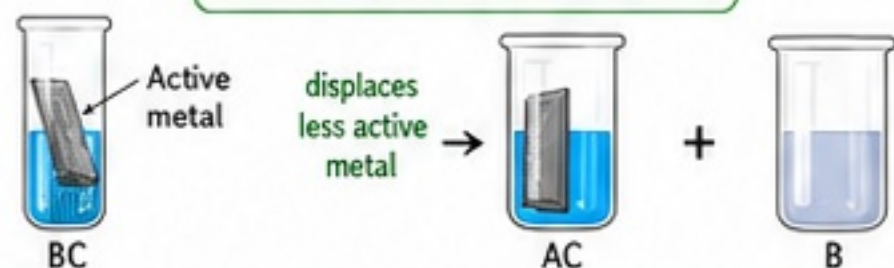
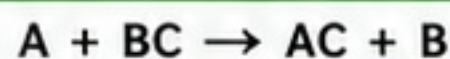


SECTION E – TYPES OF REACTIONS II: DISPLACEMENT, DOUBLE DISPLACEMENT, PRECIPITATION AND NEUTRALISATION

1 Displacement Reaction

Definition: A more reactive element displaces a less reactive element from its compound.

General Form:



Examples:

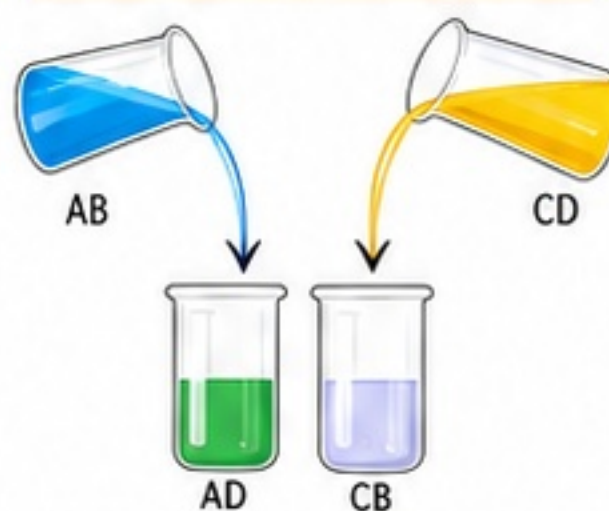
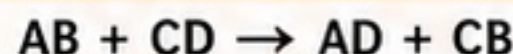
Reaction (Balanced Equation)	Observation / Colour Change
1. $\text{Zn} + \text{CuSO}_4 \text{ (blue)} \rightarrow \text{ZnSO}_4 + \text{Cu}$	Blue solution fades; Cu deposited as reddish-brown solid.
2. $\text{Fe} + \text{CuSO}_4 \text{ (blue)} \rightarrow \text{FeSO}_4 + \text{Cu}$	Blue turns pale green; Cu reddish-brown deposit.
3. $\text{Zn} + \text{FeSO}_4 \text{ (pale green)} \rightarrow \text{ZnSO}_4 + \text{Fe}$	Pale green turns colourless/silver; Fe grey deposit.

Reactivity Order (metals): $\text{K} > \text{Na} > \text{Ca} > \text{Mg} > \text{Al} > \text{Zn} > \text{Fe} > (\text{H}) > \text{Cu} > \text{Ag} > \text{Au}$

2 Double Displacement / Metathesis Reaction

Definition: Ions of two compounds exchange places to form two new compounds.

General Form:

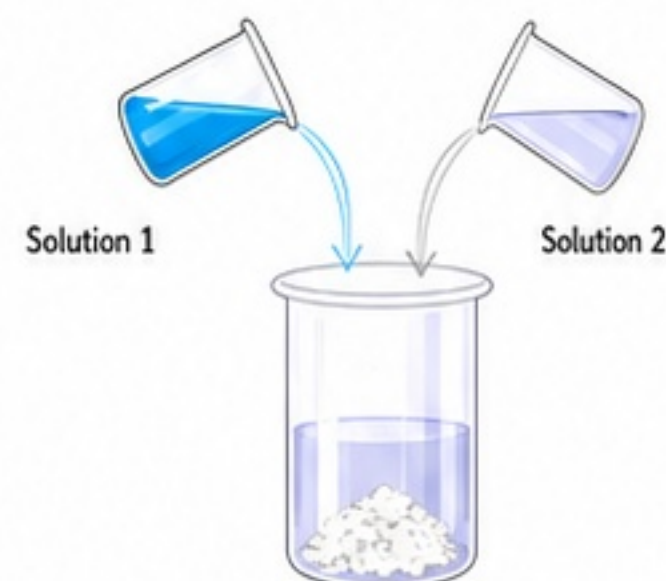


Occurs in aqueous solutions as exchange of ions.

3 Precipitation Reaction

Definition: A double displacement reaction in which an insoluble solid (precipitate) is formed.

Key Point: One of the products is an insoluble solid that separates out.



4 Important Examples of Double Displacement / Precipitation Reactions

Reaction (Balanced Equation)	Solutions (Colour)	Observation	Precipitate (Colour)	Ionic View (Net Ionic Equation)
1. $\text{AgNO}_3 \text{ (aq)} + \text{NaCl} \text{ (aq)} \rightarrow \text{AgCl} \text{ (s)} + \text{NaNO}_3 \text{ (aq)}$	AgNO_3 – colourless NaCl – colourless	White ppt appears.	AgCl – white precipitate	$\text{Ag}^+ \text{ (aq)} + \text{Cl}^- \text{ (aq)} \rightarrow \text{AgCl} \text{ (s)}$
2. $\text{BaCl}_2 \text{ (aq)} + \text{Na}_2\text{SO}_4 \text{ (aq)} \rightarrow \text{BaSO}_4 \text{ (s)} + 2\text{NaCl} \text{ (aq)}$	BaCl_2 – colourless Na_2SO_4 – colourless	White ppt appears.	BaSO_4 – white precipitate	$\text{Ba}^{2+} \text{ (aq)} + \text{SO}_4^{2-} \text{ (aq)} \rightarrow \text{BaSO}_4 \text{ (s)}$
3. $\text{Pb}(\text{NO}_3)_2 \text{ (aq)} + 2\text{KI} \text{ (aq)} \rightarrow \text{PbI}_2 \text{ (s)} + 2\text{KNO}_3 \text{ (aq)}$	$\text{Pb}(\text{NO}_3)_2$ – colourless KI – colourless	Bright yellow ppt appears.	PbI_2 – bright yellow precipitate	$\text{Pb}^{2+} \text{ (aq)} + 2\text{I}^- \text{ (aq)} \rightarrow \text{PbI}_2 \text{ (s)}$
4. $\text{CuSO}_4 \text{ (aq)} + 2\text{NaOH} \text{ (aq)} \rightarrow \text{Cu}(\text{OH})_2 \text{ (s)} + \text{Na}_2\text{SO}_4 \text{ (aq)}$	CuSO_4 – blue NaOH – colourless	Light blue ppt appears.	$\text{Cu}(\text{OH})_2$ – light blue precipitate	$\text{Cu}^{2+} \text{ (aq)} + 2\text{OH}^- \text{ (aq)} \rightarrow \text{Cu}(\text{OH})_2 \text{ (s)}$

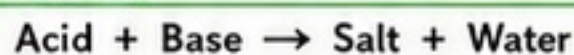
Colour Key (Solutions & Substances)

● CuSO_4 – blue
 ● FeSO_4 – pale green
 ● Cu (metal) deposit – reddish-brown
 ● NaOH – colourless solution
 ● HCl – colourless acid
 ● AgCl – white
 ● BaSO_4 – white
 ● PbI_2 – bright yellow

5 Neutralisation Reaction

Definition: Reaction between an acid and a base to form salt and water.

General Form:



Examples:

Reaction (Balanced Equation)	Observation / Indicator
1. $\text{HCl} \text{ (aq)} + \text{NaOH} \text{ (aq)} \rightarrow \text{NaCl} \text{ (aq)} + \text{H}_2\text{O} \text{ (l)}$	No colour change. NaOH colourless; HCl colourless.
2. $\text{H}_2\text{SO}_4 \text{ (aq)} + 2\text{NaOH} \text{ (aq)} \rightarrow \text{Na}_2\text{SO}_4 \text{ (aq)} + 2\text{H}_2\text{O} \text{ (l)}$	No colour change.
3. $\text{HCl} \text{ (aq)} + \text{NaOH} \text{ (aq)} + \text{Phenolphthalein} \rightarrow \text{NaCl} \text{ (aq)} + \text{H}_2\text{O} \text{ (l)}$	Pink in base (NaOH) turns colourless at neutrality.

6 How to Identify Each Reaction Type

Reaction Type	How to Identify
Displacement	A more reactive element replaces a less reactive element from its compound.
Double Displacement	Two compounds exchange ions; no element gets displaced.
Precipitation	Double displacement where an insoluble solid (precipitate) forms.
Neutralisation	Acid + Base $\rightarrow$ Salt + Water. pH moves towards 7. Indicator change may occur.

7 Real-Life Links



Displacement: Used in extraction of metals (e.g., aluminium extracted by displacing aluminium from alumina).



Precipitation: Water purification – removal of harmful ions by forming insoluble precipitates.



Neutralisation: Antacids neutralise excess stomach acid (HCl).



Double Displacement: Used in making medicines, salts, fertilizers and in chemical industries.

8 Quick Memory Cues



Displacement: "More Active takes the place" (A displaces B).



Precipitation: "Look for a solid – it's precipitating!"



Neutralisation: "Acid + Base $\rightarrow$ Salt + Water" (Always watery!).



Double Displacement: "Swap and form two new combinations."

9 Exam Tips / Deep Points

- ★ Balance all chemical equations correctly.
- ★ Write states (s), (l), (aq), (g) in every equation.
- ★ In precipitation, remember the colour of the precipitate.
- ★ Use solubility rules: All nitrates and sodium salts are soluble.
- ★ For displacement, remember reactivity series.
- ★ In neutralisation with phenolphthalein: Pink in base $\rightarrow$ colourless at neutral.
- ★ Write net ionic equation for precipitation reactions.

10 At a Glance Summary

Type	Key Feature
Displacement	Element displaces another from compound. $A + BC \rightarrow AC + B$
Double Displacement	Ions exchange between two compounds. $AB + CD \rightarrow AD + CB$
Precipitation	Double displacement producing insoluble solid (precipitate).
Neutralisation	Acid + Base $\rightarrow$ Salt + Water. pH approaches 7.



Ch 1: Chemical Reactions and Equations

Part 1 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

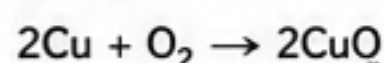


SECTION F – OXIDATION, REDUCTION AND REDOX REACTIONS

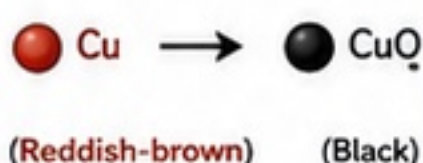
1 Oxidation

Oxidation is the process in which a substance loses electrons or gains oxygen or loses hydrogen.

Example:



Cu is oxidised.



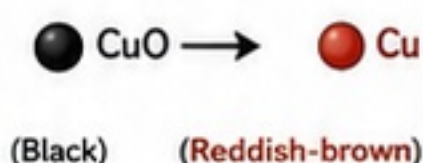
2 Reduction

Reduction is the process in which a substance gains electrons or loses oxygen or gains hydrogen.

Example:



CuO is reduced.



3 Why They Occur Together

In a redox reaction, the electrons lost by one substance are gained by another. So, oxidation and reduction occur simultaneously.

Remember:

Oxidation = Loss of e^-

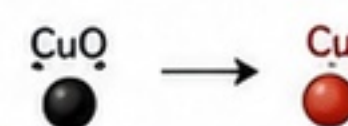
Reduction = Gain of e^-

4 Oxidising Agent

The substance that causes oxidation by gaining electrons is called the oxidising agent. It gets reduced in the process.

Example:

In $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, CuO is the oxidising agent (Black).

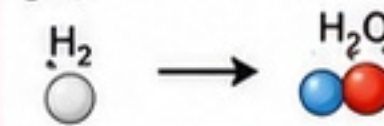


5 Reducing Agent

The substance that causes reduction by losing electrons is called the reducing agent. It gets oxidised in the process.

Example:

In $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, H_2 is the reducing agent (Colourless gas).



6 Key Idea

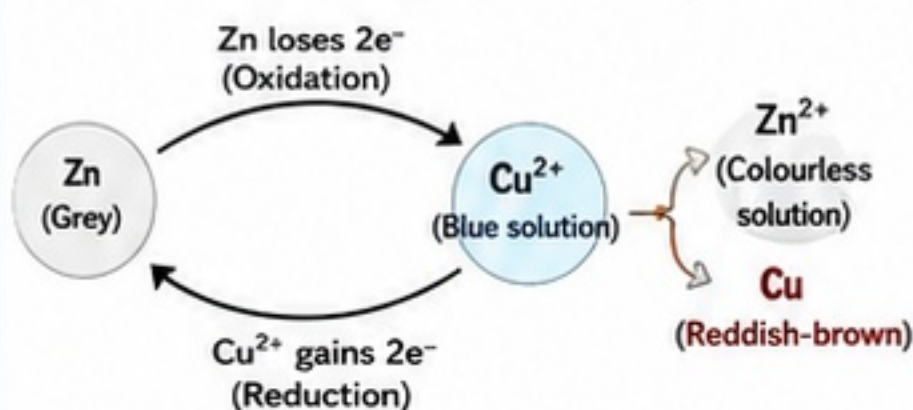
One substance is oxidised, the other is reduced. Oxidising agent gets reduced. Reducing agent gets oxidised.



7 Examples of Redox Reactions

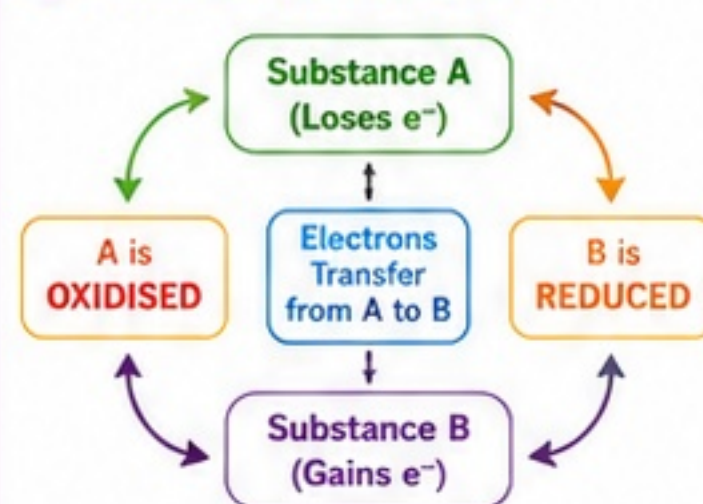
S. No.	Balanced Reaction	What is Oxidised (Loses e^-)	What is Reduced (Gains e^-)	Oxidising Agent	Reducing Agent	Observation / Colour Change
1	$2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$	Cu (Reddish-brown)	O_2 (Colourless gas)	O_2	Cu	Copper turns black (CuO black forms).
2	$\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$	H_2 (Colourless gas)	CuO (Black)	CuO	H_2	Black CuO changes to reddish-brown Cu; water (colourless liquid) forms.
3	$\text{Fe} + \text{Cl}_2 \rightarrow \text{FeCl}_3$	Fe (Grey)	Cl_2 (Greenish-yellow gas)	Cl_2	Fe	Greenish-yellow Cl_2 gas reacts; brown fumes of FeCl_3 form.
4	$2\text{Fe} + 3\text{Cl}_2 \rightarrow 2\text{FeCl}_3$	Fe (Grey)	Cl_2 (Greenish-yellow gas)	Cl_2	Fe	Hot reaction with greenish-yellow Cl_2 ; reddish-brown fumes of FeCl_3 .
5	$\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$	Zn (Grey)	Cu^{2+} in CuSO_4 (Blue solution)	CuSO_4 (Blue)	Zn	Blue CuSO_4 changes to colourless ZnSO_4 ; reddish-brown Cu deposits on Zn.
6	$4\text{Fe} + 3\text{O}_2 + x\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (Rust formation)	Fe (Grey)	O_2 (Colourless gas)	O_2	Fe	Rust ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$) forms (Reddish-brown).

8 Electron-Transfer Diagram (Example)



Electrons lost by Zn are gained by Cu^{2+} . Oxidation and Reduction occur together.

9 Redox Flow Diagram



Oxidation and Reduction always occur simultaneously.

10 Redox in Everyday Life

-  **Rusting of iron:** Iron reacts with O_2 and water vapour to form rust ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, reddish-brown).
-  **Respiration:** Glucose is oxidised in our cells to release energy; O_2 is reduced to H_2O .
-  **Corrosion prevention:** Painting, oiling, galvanisation (Zn coating) prevent oxidation (rusting).
-  **Bleaching:** H_2O_2 (colourless liquid) is oxidised to H_2O and helps remove stains.
-  **Disinfection:** KMnO_4 (purple) and Cl_2 (greenish-yellow gas) act as strong oxidising agents.
-  **Iodine clock reaction:** I^- is oxidised to I_2 (brown solution or violet vapour).

11 Key Takeaways

- ✓ Oxidation = Loss of electrons / Gain of oxygen / Loss of hydrogen.
- ✓ Reduction = Gain of electrons / Loss of oxygen / Gain of hydrogen.
- ✓ Oxidising agent gets reduced.
- ✓ Reducing agent gets oxidised.
- ✓ Electrons lost by one substance are gained by another.
- ✓ Redox reactions are very common in nature and daily life.

12 Memory Trick

OIL RIG

O – Oxidation is Loss of electrons
I – Increase in oxygen
L – Loss of hydrogen

R – Reduction is Gain of electrons
I – Increase in hydrogen
G – Gain of oxygen



13 Exam Tip

- ★ Always identify what is oxidised and what is reduced.
- ★ Check electron balance to confirm redox reaction.
- ★ Remember typical colours:

CuO – Black	Cu – Reddish-brown	Fe_2O_3 – Reddish-brown
CuSO_4 – Blue	ZnSO_4 – Colourless	MgO – White
KMnO_4 – Purple	Cl_2 – Greenish-yellow gas	
I_2 – Brown solution or Violet vapour		
H_2O_2 – Colourless liquid	Rust – Reddish-brown	
- ★ Write observations in exams with colour changes.





Ch 1: Chemical Reactions and Equations

Part 1 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus



SECTION G – EXOTHERMIC AND ENDOTHERMIC REACTIONS

1 Key Idea: Energy Change

In every chemical reaction, energy is either released or absorbed.

- **Exothermic reaction:** Energy is released to the surroundings (as heat).
- **Endothermic reaction:** Energy is absorbed from the surroundings.

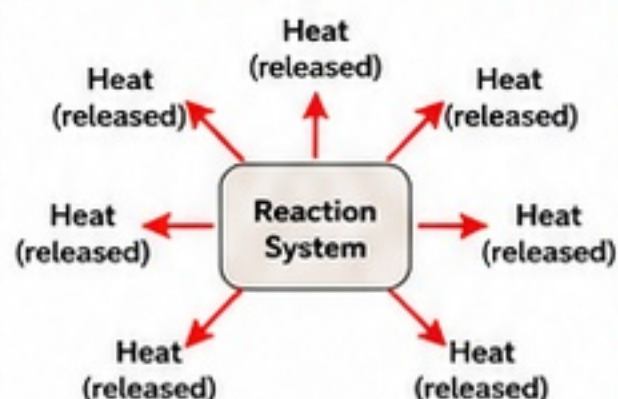
Remember:

Energy is neither created nor destroyed. It only changes form.

2 Exothermic Reactions

Reactions that release heat energy to the surroundings.
Temperature of surroundings increases.
 ΔH (change in enthalpy) is negative.

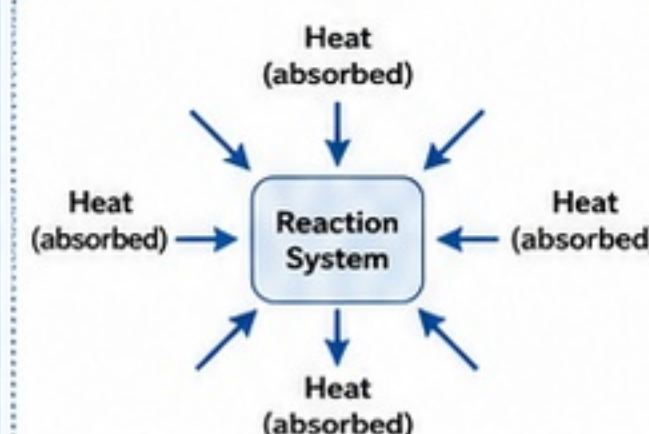
Heat-Flow Diagram



3 Endothermic Reactions

Reactions that absorb heat energy from the surroundings.
Temperature of surroundings decreases.
 ΔH (change in enthalpy) is positive.

Heat-Flow Diagram

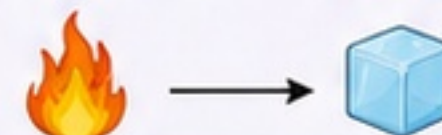


4 Why Does Energy Change?

- Breaking bonds requires energy (input).
- Forming bonds releases energy (output).
- If energy released > energy absorbed → Exothermic
- If energy absorbed > energy released → Endothermic

Key Point:

Bond making releases energy.
Bond breaking requires energy.



5 Examples in Daily Life

Exothermic Reactions (Heat is Released)

Example & Equation	Observation
 1. Burning of Magnesium $2\text{Mg (s)} + \text{O}_2 \text{ (g)} \rightarrow 2\text{MgO (s)}$ (White magnesium oxide)	 Burns with dazzling white light and gives off heat. White ash (MgO) is left.
 2. Combustion of Methane $\text{CH}_4 \text{ (g)} + 2\text{O}_2 \text{ (g)} \rightarrow \text{CO}_2 \text{ (g)} + 2\text{H}_2\text{O (l)}$ (carbon dioxide, colourless) (water, colourless)	 Burns with blue flame. Heat and light are released.
 3. Slaking of Quicklime $\text{CaO (s)} + \text{H}_2\text{O (l)} \rightarrow \text{Ca(OH)}_2 \text{ (aq)}$ (calcium hydroxide solution)	 Large amount of heat is released. Mixture becomes hot.

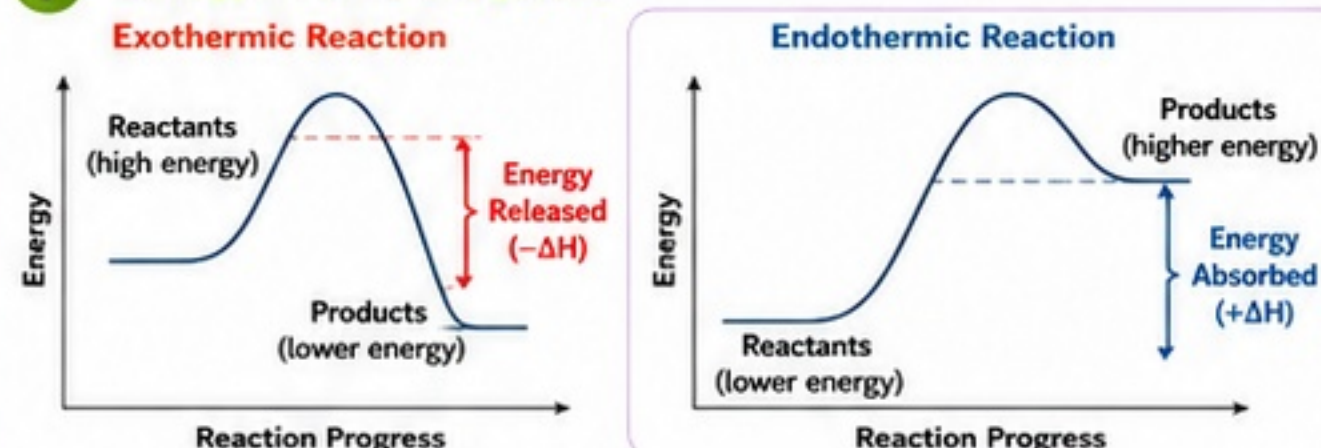
Endothermic Reactions (Heat is Absorbed)

Example & Equation	Observation
1. Thermal Decomposition of Calcium Carbonate $\text{CaCO}_3 \text{ (s)} \text{ (white)} \rightarrow \text{CaO (s)} \text{ (white)} + \text{CO}_2 \text{ (g)}$ (carbon dioxide, colourless)	 Test tube becomes hot but reaction needs heat. CO_2 gas turns lime water milky.
2. Dissolution of Ammonium Nitrate $\text{NH}_4\text{NO}_3 \text{ (s)} + \text{H}_2\text{O (l)} \rightarrow \text{NH}_4\text{NO}_3 \text{ (aq)}$ (colourless solution)	 Beaker feels cold. Temperature of solution decreases. Solution remains colourless.
3. Dissolution of Copper(II) Sulphate $\text{CuSO}_4 \text{ (s)} \text{ (blue crystals)} + 5\text{H}_2\text{O (l)} \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O (aq)}$ (blue solution)	 Blue crystals dissolve and absorb heat. Beaker feels cool.

6 Comparison between Exothermic and Endothermic Reactions

Feature	Exothermic Reactions	Endothermic Reactions
Energy Change	Release energy to surroundings	Absorb energy from surroundings
Sign of ΔH	Negative ($\Delta H < 0$)	Positive ($\Delta H > 0$)
Effect on Surroundings	Temperature increases	Temperature decreases
Heat-Flow	Heat flows out of system	Heat flows into system
Examples	Combustion, slaking of lime, neutralisation	Thermal decomposition, dissolving NH_4NO_3 , photosynthesis
Common Analogy	Hand warmers, burning candle	Instant cold packs, ice melt

7 Energy Profile Diagrams



8 Where We See Them in Daily Life

 Exothermic Examples	 Endothermic Examples
• Burning of fuels (LPG, petrol) for cooking. • Hand warmers. • Reaction of acid with base (neutralisation). • Slaking of lime in cement manufacture.	• Instant cold packs. • Photosynthesis in plants. • Thermal decomposition in baking soda. • Dissolving NH_4NO_3 in ice packs.

9 Practical Applications

 Heating	Exothermic reactions are used in heating pads, fuel cells, rocket fuels.
 Cooling	Endothermic reactions are used in instant cold packs, refrigeration.
 Industries	Exothermic combustion provides energy for power plants.
 Agriculture	Photosynthesis (endothermic) helps plants grow using sunlight.

10 Exam Tips

- ★ Remember: Energy change is always part of a reaction.
- ★ Exothermic = Heat Out (Temperature ↑), $\Delta H < 0$.
- ★ Endothermic = Heat In (Temperature ↓), $\Delta H > 0$.
- ★ Learn energy profile diagrams well.
- ★ Write balanced equations with states.
- ★ Mention observations: colour, temperature, gas evolved.
- ★ Use key words like "releases heat" or "absorbs heat" in answers.



11 Quick Key Terms

Term	Definition
Exothermic Reaction	Reaction that releases heat to the surroundings.
Endothermic Reaction	Reaction that absorbs heat from the surroundings.
Enthalpy Change (ΔH)	Change in heat content; negative for exothermic, positive for endothermic reactions.
Activation Energy	Minimum energy required for a reaction to start.

12 Quick Recall

- ★ Exothermic: Heat released, surroundings warm, $\Delta H < 0$, energy of products < reactants.
- ★ Endothermic: Heat absorbed, surroundings cool, $\Delta H > 0$, energy of products > reactants.
- ★ Burning $\text{Mg} \rightarrow \text{MgO}$ (white) + heat (exothermic).
- ★ CaCO_3 (white) $\rightarrow$ CaO (white) + CO_2 (colourless) needs heat (endothermic).
- ★ NH_4NO_3 dissolves $\rightarrow$ solution colourless, beaker feels cold (endothermic).





Ch 1: Chemical Reactions and Equations

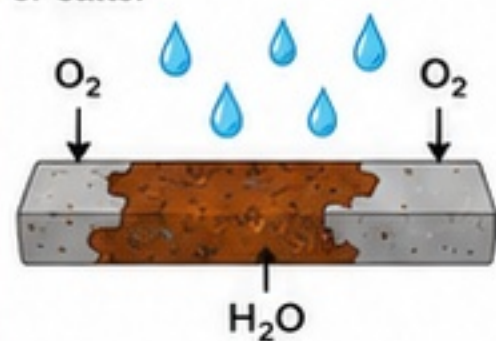
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SECTION H – CORROSION

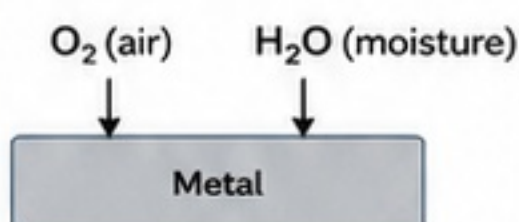
1 What is Corrosion?

Corrosion is the gradual destruction of metals by chemical reactions with their environment (air, moisture, acids, etc.). It is a natural process that returns metals to their more stable compounds such as oxides, hydroxides or salts.



2 Why Corrosion Happens?

- Metals are in a high-energy state and tend to revert to more stable compounds.
- Presence of air (oxygen), moisture and sometimes acids/salts speeds up the reaction.
- Electrochemical reactions occur on the metal surface forming oxides or other compounds.



3 Rusting of Iron Explained

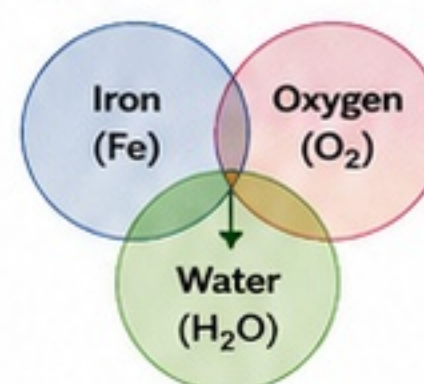
- Iron reacts slowly with oxygen and water present in the air to form rust.
- Rust is a reddish-brown hydrated iron(III) oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$).
- Rust is porous and flaky, so it does not protect iron. The reaction continues and more iron gets destroyed.



4 Conditions Necessary for Rusting

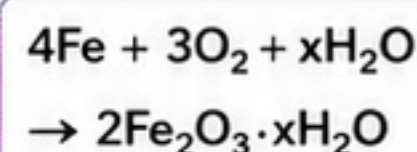
All the following are necessary for rusting of iron:

- Presence of iron
- Presence of oxygen (air)
- Presence of water (moisture)



5 Equation for Rust Formation

Overall reaction (simplified):



Where,

$\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ = Rust (reddish-brown hydrated iron(III) oxide)

Rust colour:
Reddish-brown

6 Real-Life Examples of Corrosion (with Colour of Products)

- Iron (steel structures, gates, railings): Forms rust – reddish-brown hydrated iron(III) oxide.
- Copper articles (statues, roofs): Form green patina – basic copper carbonate [$\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$]. Colour: Green.
- Silver articles (jewellery, cutlery): Get black tarnish – silver sulphide (Ag_2S). Colour: Black.
- Aluminium: Forms a thin, dull grey protective layer of aluminium oxide (Al_2O_3) that protects further corrosion. Colour: Thin dull grey.
- Zinc: Forms a silvery coating of zinc carbonate / zinc oxide (ZnCO_3 , ZnO). Colour: Silvery white.
- Chromium: Forms a shiny protective layer of chromium oxide (Cr_2O_3). Colour: Shiny.

7 Disadvantages of Corrosion

- Loss of valuable metals due to wastage.
- Weakening and damage of structures and machines.
- Leakage of pipes and tanks.
- Shortened life of vehicles, bridges, tools, etc.
- High maintenance and repair costs.
- Affects safety and can cause accidents.



8 Prevention of Corrosion – Comparison Table

Method	Painting	Oiling / Greasing	Galvanisation	Alloying	Electroplating
How it Works	A protective layer of paint prevents contact of metal with air and moisture.	A thin layer of oil or grease blocks air and moisture.	Coating iron/steel with zinc forms a protective barrier. Zinc is more reactive and protects iron even if coating gets scratched.	Adding other metals (e.g., Cr, Ni) forms alloys that resist corrosion.	A thin layer of another metal (e.g., Cr, Ni, Zn) is deposited on the surface using electric current.
Cross-section Diagram					
Example	Painting iron gates, railings, machines.	Oiling machine parts, bicycle chains.	Galvanised iron sheets, GI pipes.	Stainless steel (Fe + Cr + Ni), Nichrome.	Chromium plating on taps, nickel plating on cutlery.
Colour of Protective Layer / Coating	As per paint used (e.g., red, blue).	Transparent / yellowish.	Silvery (zinc coating).	Silvery / metallic (shiny).	Depends on metal: Cr – Shiny silvery Ni – Silvery white Zn – Silvery

9 Observation Notes – Rusting Experiment

Experiment: Iron nail in different conditions

	Dry air (CaCl_2)	Boiled water (no air)	Tap water (with air)	Salt water (with air)
Observation:	No rust	No rust	Rust forms (slow)	Rust forms (fastest)
Inference:	Both air and water are needed.	No air, so no rust.	Both air and water present → rust.	Salt water speeds up rusting.

10 Daily-Life Applications

- Protective coatings increase the life of bridges, buildings, vehicles and tools.
- Galvanised sheets used in roofs and water tanks.
- Stainless steel used in kitchen sinks, appliances and medical instruments.
- Electroplated parts used in taps, cycle parts, car parts for better look and durability.
- Rust converters and primers used before painting metal surfaces.

11 Quick Recall

- ★ Corrosion = Destruction of metals by chemical reactions with environment.
- ★ Rust = Reddish-brown hydrated iron(III) oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$).
- ★ Conditions for rusting: Iron + Oxygen + Water (all three are necessary).
- ★ Copper patina = Green.
- ★ Silver tarnish = Black.
- ★ Aluminium oxide = Thin dull grey (protective).
- ★ Zinc coating = Silvery.
- ★ Chromium oxide = Shiny protective layer.
- ★ Best prevention methods: Barrier coatings, protective coatings, and using suitable alloys.





Ch 1: Chemical Reactions and Equations

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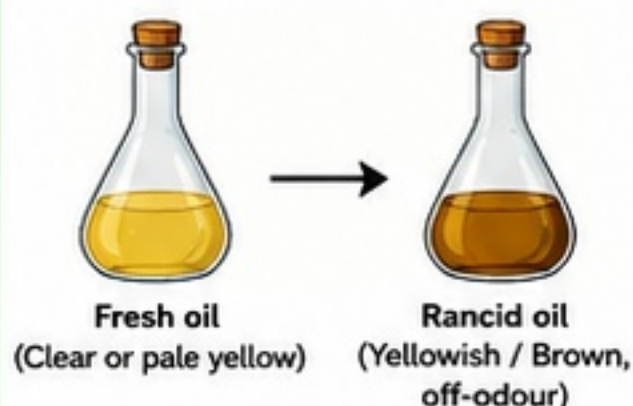
SECTION I – RANCIDITY AND REAL-LIFE APPLICATIONS

1 What is Rancidity?

Rancidity is the oxidation of fats and oils present in food, which produces unpleasant smell and taste, and makes them unfit for consumption.

Observation:

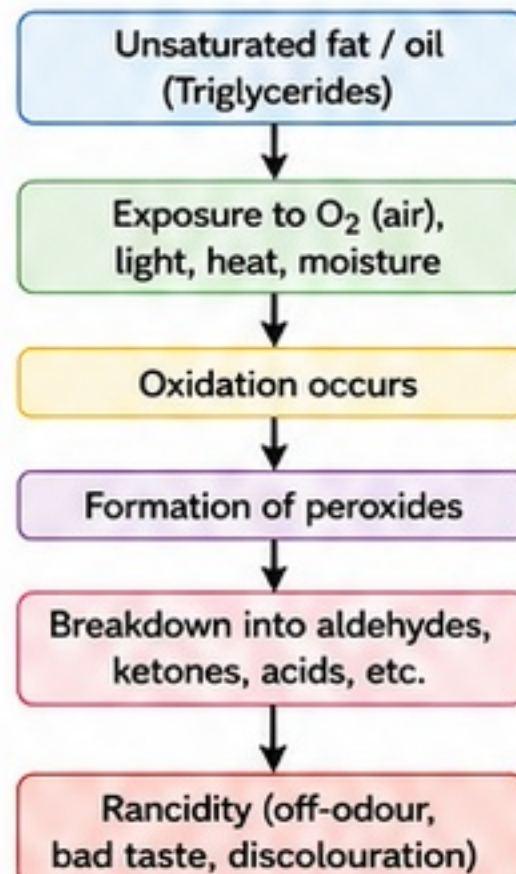
Rancid fats and oils often become **yellowish, brown or darken** and develop an **off-odour**.



2 Why Fats and Oils Become Rancid

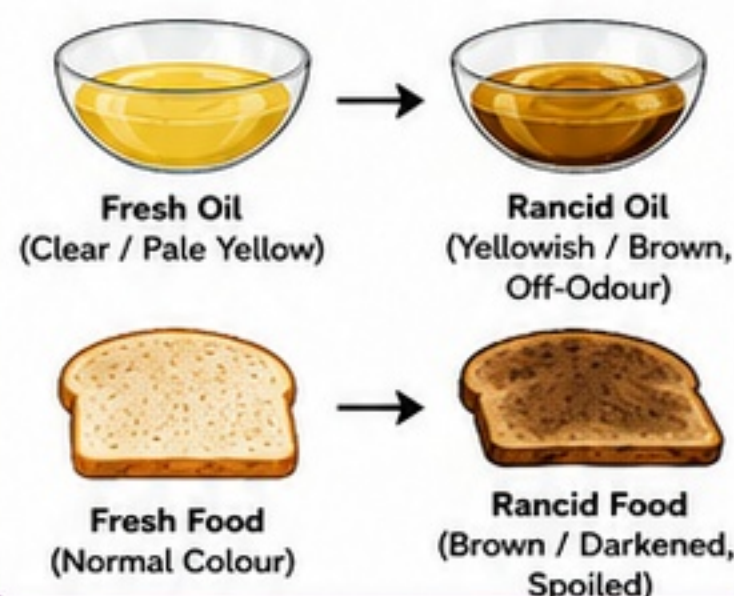
- Fats and oils contain unsaturated fatty acids.
- When exposed to air (oxygen), light, heat or moisture for a long time, they undergo oxidation.
- The oxidation breaks the molecules and forms peroxides, aldehydes and other compounds.
- These products cause off-odour and unpleasant taste (rancidity).

3 Oxidation Behind Rancidity (Flow Diagram)



4 Signs of Rancidity (Observations)

- Yellowish colour develops in oils.
- Change to brown or dark colour in food.
- Unpleasant, stale, or off-odour.
- Bitter or unpleasant taste.
- Formation of free fatty acids.
- Food becomes unfit for use.



5 Methods to Prevent Rancidity

- Store oils and fats in airtight containers to reduce contact with oxygen.
- Keep away from light and heat.
- Add antioxidants (e.g., BHA, BHT, Vitamin E) to delay oxidation.
- Refrigerate to slow down oxidation.
- Use nitrogen flushing (colourless gas) in packaging to displace oxygen.

6 Examples of Rancidity in Daily Life

- Cooking oils (groundnut, sunflower, etc.) left open turn yellowish and give off a stale smell.
- Fried snacks like chips become stale and taste bad after some time.
- Butter and ghee develop an off-odour and taste when stored for long.
- Nuts and seeds become oily, darken and smell bad when rancid.
- Bakery products (cakes, biscuits) may turn brown and develop bad odour when fats go rancid.

7 Corrosion vs Rancidity

Aspect	Corrosion	Rancidity
What it affects	Metals (iron, copper, etc.)	Fats and oils in food
Cause	Reaction with air, moisture, acids (Chemical reaction)	Oxidation by oxygen in air (Chemical reaction)
Products	Metal oxides / hydroxides (e.g., rust)	Peroxides, aldehydes, ketones, free fatty acids, etc.
Observation	Rust (reddish brown coating)	Yellowish / brown colour, off-odour, bad taste
Effect	Weakens and damages metal	Food becomes spoiled and unfit to eat
Prevention	Painting, oiling, galvanization, alloying	Airtight storage, refrigeration, antioxidants, nitrogen packaging

8 Importance of Chemical Reactions in Real Life

Combustion / Cooking - Fuels (LPG, wood, coal) burn in oxygen to produce heat and energy. - Food is cooked by chemical reactions (browning, baking, frying). - Example: Burning of LPG produces heat.	Digestion Food is broken down into simpler substances by chemical reactions in our stomach. - Enzymes help in these reactions to release energy and nutrients. - Example: Carbohydrates → simple sugars.	Respiration Glucose reacts with oxygen in cells to release energy. - Produces carbon dioxide, water and energy. - Example: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$	Industry / Manufacturing Chemical reactions are used to make medicines, soaps, fertilizers, plastics, glass, paper, etc. - Large-scale chemical reactions help in producing useful materials. - Example: Haber process for ammonia (fertilizers).	Environment - Photosynthesis: Plants make food using sunlight, CO_2 and water. - Oxygen released is used by living organisms. - Example: $CO_2 + H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + O_2$
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9 One-Line Recall

- Rancidity is the oxidation of fats and oils causing off-odour and bad taste.
- Oxygen (colourless gas) causes fats and oils to turn rancid.
- Fresh oils are pale yellow; rancid oils turn yellowish or brown.
- Rancid food develops off-odour and darkening.
- Antioxidants and nitrogen gas (colourless) help prevent rancidity.

10 Exam Tips / Quick Recall

- Remember: Unsaturated fats + $O_2 \rightarrow$ Oxidation $\rightarrow$ Rancidity.
- Observe colour change (yellowish / brown) and off-odour.
- Corrosion affects metals; rancidity affects food fats/oils.
- Store oils in airtight containers, in cool and dark places.
- Learn real-life applications: cooking, digestion, respiration, industry, environment.

11 High-Yield Exam Pointers

- ✓ Definition + cause + signs of rancidity.
- ✓ Prevention methods (antioxidants, nitrogen packaging, refrigeration).
- ✓ Corrosion vs rancidity comparison.
- ✓ Real-life applications of chemical reactions.
- ✓ Flow diagram of oxidation leading to rancidity.
- ✓ Observations: yellowish oils, brown food, off-odour, colourless O_2 and N_2 .



Ch 1: Chemical Reactions and Equations



Part 1 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION J – MASTER REVISION AND EXAM SHEET

1 Master Revision – Part 1 at a Glance

- ✓ Substances react to form new substances.
- ✓ Reactants → Products with different properties.
- ✓ Atoms are rearranged; not created or destroyed.
- ✓ Mass is conserved in a chemical reaction.
- ✓ Chemical equations use symbols and formulae.
- ✓ Equations are balanced to obey the law of conservation of mass.
- ✓ Reactions are identified by signs like change in state, colour, temperature, gas, precipitate, smell.
- ✓ Many reactions are useful in daily life.

2 Most Important Definitions

- **Chemical Reaction:** A process in which one or more substances (reactants) change into new substances (products) with different properties.
- **Reactants:** The starting substances in a reaction.
- **Products:** The new substances formed.
- **Balanced Chemical Equation:** An equation in which the number of atoms of each element is the same on both sides.
- **Law of Conservation of Mass:** Mass can neither be created nor destroyed in a chemical reaction.
- **Precipitate:** An insoluble solid formed in a reaction.

3 Major Types of Chemical Reactions

Type of Reaction	General Equation	Example (Balanced)	Key Feature	Observation / Colour (if relevant)
Combination (Synthesis)	$A + B \rightarrow AB$	$2Mg + O_2 \rightarrow 2MgO$	Two or more reactants form one product.	MgO forms white ash.
Decomposition	$AB \rightarrow A + B$	$CaCO_3 \rightarrow CaO + CO_2$	One compound breaks into two or more.	CO_2 gas evolves; CaO is white.
Displacement (Single)	$A + BC \rightarrow AC + B$	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$	More reactive element displaces less reactive one.	$CuSO_4$ blue changes to colourless $ZnSO_4$; Cu reddish-brown deposits.
Double Displacement (Metathesis)	$AB + CD \rightarrow AD + CB$	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$	Ions exchange partners to form new compounds.	$BaSO_4$ is white precipitate.
Precipitation	Soluble + Soluble → Insoluble + Soluble	$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$	An insoluble solid (precipitate) forms.	AgCl is white precipitate.
Neutralisation (Acid + Base)	Acid + Base → Salt + Water	$HCl + NaOH \rightarrow NaCl + H_2O$	Acid and base react to form salt and water.	No colour change; solution becomes neutral.

4 10 Must-Remember Balanced Equations (With Observations / Colour Changes)

1. $2H_2 + O_2 \rightarrow 2H_2O$	Water forms.
2. $2Mg + O_2 \rightarrow 2MgO$	MgO white ash.
3. $CaCO_3 \rightarrow CaO + CO_2$	CO_2 gas evolves.
4. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$	Cu reddish-brown; $CuSO_4$ blue → colourless.
5. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$	Cu reddish-brown; $FeSO_4$ pale green solution.
6. $2AgNO_3 + 2NaCl \rightarrow 2AgCl + 2NaNO_3$	AgCl white precipitate.
7. $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$	$BaSO_4$ white precipitate.
8. $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$	PbI_2 yellow precipitate.
9. $4Fe + 3O_2 + 6H_2O \rightarrow 4Fe(OH)_3$	Rust ($Fe(OH)_3$) forms reddish-brown.
10. $HCl + NaOH \rightarrow NaCl + H_2O$	Salt and water form; no colour change.

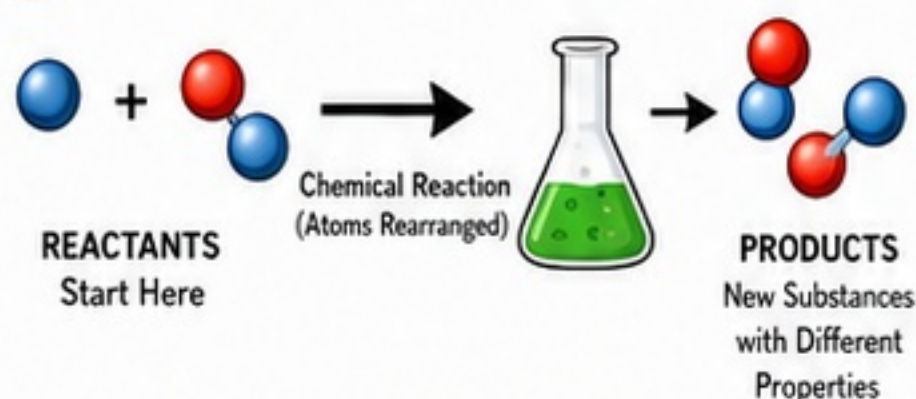
5 Common Mistakes Students Make

- ✗ Not balancing the equation properly.
- ✗ Forgetting physical states (s), (l), (g), (aq).
- ✗ Writing formulas incorrectly (e.g., CaO instead of CO_2).
- ✗ Ignoring small coefficients and writing them as 1.
- ✗ Not correctly identifying reactants and products.
- ✗ Ignoring observations and colour changes.
- ✗ Thinking mass is created or destroyed.
- ✗ Mixing up similar salts (e.g., $CuSO_4$ vs $FeSO_4$).
- ✗ Not checking atom count after balancing.

6 Exam-Winning Tips

- ★ Always balance equations step-by-step.
- ★ Write physical states and conditions wherever needed.
- ★ Identify type of reaction by pattern.
- ★ Memorise key observations and colours.
- ★ Count atoms of each element on both sides to check balance.
- ★ Use box method or inspection method.
- ★ Read the question carefully – identify what is given and what is asked.
- ★ Draw rough diagrams where needed.
- ★ Write neat, labelled answers – presentation matters!

7 Quick Visual Recap – Reactants to Products



8 Law of Conservation of Mass



9 Formula & Equations Quick Reference

- ★ General Equation: $aA + bB \rightarrow cC + dD$
- ★ Mass is conserved in all reactions.
- ★ Physical states: (s) solid, (l) liquid, (g) gas, (aq) aqueous
- ★ Arrow (→) reads as "yields" or "produces"
- ★ Coefficients balance the number of atoms.

10 Checklist – What to Revise Before Exam

- ☐ Types of chemical reactions
- ☐ Balancing of equations
- ☐ Signs / Indicators of reactions
- ☐ Key observations & colours
- ☐ Important equations (at least 10)
- ☐ Law of conservation of mass
- ☐ Definitions & terminology
- ☐ Important diagrams & examples

11 Self-Check: Performance Tracker

Topic	Confident (✓)	Need More Practice (?)	Not Yet (X)
Chemical Reaction – Definition			
Reactants & Products			
Law of Conservation of Mass			
Types of Chemical Reactions			
Balancing Chemical Equations			
Signs / Indicators of Reaction			
Colour Changes & Observations			
Real-Life Examples			
Important Equations (10)			
Overall Understanding			

12 Challenge Yourself – Quick Recall

1. What is a chemical reaction?
2. Why do we balance chemical equations?
3. State the law of conservation of mass.
4. Write the balanced equation for: (a) $Zn + CuSO_4$ (b) $AgNO_3 + NaCl$
5. What colour change is observed when iron reacts with $CuSO_4$ solution?
6. Name the type of reaction: $CaCO_3 \rightarrow CaO + CO_2$
7. What is the colour of $FeSO_4$ solution?
8. What is the colour of rust?
9. Write the balanced equation for formation of rust.
10. Give two examples of precipitation reactions with colours.

Practice Today, Score Centum Tomorrow!



CHEMISTRY IS NOT JUST A SUBJECT – IT'S THE REACTION THAT CREATES YOUR SUCCESS!





Ch 1: Chemical Reactions and Equations

Part 2 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus



SECTION K – MASTER COLOUR AND OBSERVATION ATLAS

1 Why Colours and Observations Matter

- Colours and changes are the first signs of a chemical reaction.
- They help us identify reactants, products, gases and precipitates.
- Observations support balanced equations and confirm that a reaction has taken place.
- In exams, correct colours and observations fetch easy marks!

2 Common Solids / Metals / Non-metals

Substance	Appearance / Colour
Cu	Reddish-brown metal
Fe	Grey metal
Zn	Grey metal
Ag	Lustrous grey metal
Hg	Silvery liquid (metal)
S	Yellow solid
C	Black solid
Mg	Silvery white metal
Al	Silvery white metal

3 Common Solutions and Their Colours

Solution	Colour
CuSO ₄ solution	Blue
FeSO ₄ solution	Pale green
ZnSO ₄ solution	Colourless
NaCl solution	Colourless
NaOH solution	Colourless
HCl solution	Colourless
AgNO ₃ solution	Colourless
Pb(NO ₃) ₂ solution	Colourless
BaCl ₂ solution	Colourless

4 Common Solids / Products and Their Colours

Compound / Product	Colour / Appearance
CuO	Black solid
Fe ₂ O ₃	Reddish-brown solid
Fe ₂ O ₃ · xH ₂ O (Rust)	Reddish-brown
MgO	White ash
CaO	White solid
AgCl	White precipitate
BaSO ₄	White precipitate
PbI ₂	Bright yellow precipitate
Cu	Reddish-brown deposit
MnO ₂	Black solid
CaCO ₃	White solid
Cu(OH) ₂	Light blue precipitate

5 Common Gases and Their Appearance / Test

Gas	Appearance / Colour / Test
H ₂	Colourless gas; burns with 'pop' sound
O ₂	Colourless gas; rekindles glowing splint
CO ₂	Colourless gas; turns lime water milky
Cl ₂	Greenish-yellow gas; bleaches moist litmus paper
HCl	Colourless gas; pungent smell; forms white fumes with NH ₃

6 Observation-Based Reactions

Reaction	Balanced Equation	Observation / Colour Change
Iron + Copper(II) sulphate	Fe(s) + CuSO ₄ (aq) → FeSO ₄ (aq) + Cu(s)	Blue solution turns pale green; reddish-brown copper deposit forms.
Silver nitrate + Sodium chloride	AgNO ₃ (aq) + NaCl(aq) → AgCl(s)↓ + NaNO ₃ (aq)	White precipitate (AgCl) forms.
Lead(II) nitrate + Potassium iodide	Pb(NO ₃) ₂ (aq) + 2KI(aq) → PbI ₂ (s)↓ + 2KNO ₃ (aq)	Bright yellow precipitate (PbI ₂) forms.
Calcium carbonate (heating)	CaCO ₃ (s) → CaO(s) + CO ₂ (g)	White solid turns to white ash; colourless gas evolves; turns lime water milky.
Copper + Oxygen (heating)	2Cu(s) + O ₂ (g) → 2CuO(s)	Reddish-brown copper turns black.

7 Quick Memory Cues

- Blue** is for **Copper** – CuSO₄ (blue), Cu(OH)₂ (light blue), Cu is reddish-brown.
- Green** for **Iron** – FeSO₄ (pale green), rust is reddish-brown.
- Yellow** for **Lead Iodide** – PbI₂ is bright yellow.
- White** is common – many precipitates (AgCl, BaSO₄, CaCO₃) are white.
- Black** indicates oxides – CuO, MnO₂, C (carbon).

8 Exam Tips

- Always write the colour/appearance of solutions, precipitates, gases and products.
- Mention observations in full sentences.
- Use precise words: colourless, pale green, bright yellow, white precipitate, black solid, etc.
- Do not write "ppt" or "gas" without describing colour/appearance.

9 One-Minute Recall

- Solutions:** CuSO₄ blue; FeSO₄ pale green; others colourless.
- Precipitates:** AgCl white; BaSO₄ white; PbI₂ bright yellow; Cu(OH)₂ light blue.
- Gases:** H₂ pop; O₂ rekindles; CO₂ milky lime water; Cl₂ greenish-yellow; HCl pungent.
- Key changes** show reactions – colour change, precipitate, gas, heat/light.

10 More Observation Points

- Change in colour of solution.
- Formation of precipitate (white, yellow, blue, etc.).
- Evolution of gas (bubbles, smell, test).
- Change in temperature (heat evolved or absorbed).
- Change in state (solid → gas, solid → solid, solution → precipitate).



Ch 1: Chemical Reactions and Equations

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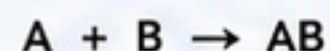


SECTION L – COMBINATION AND DECOMPOSITION REACTIONS (DEEP STUDY)

1 Combination Reaction

Two or more substances combine to form a single product.

General Form



Key Points

- Two or more reactants.
- Only one product is formed.
- Usually involves release of heat.
- Often results in increase in mass.

2 Decomposition Reaction

A single compound breaks down to form two or more simpler substances.

General Form



Key Points

- One reactant (compound).
- Two or more products formed.
- Requires energy (heat, electricity or light).
- Often results in increase in volume.

3 Types of Decomposition Reactions

Thermal Decomposition

Decomposition caused by heat.

Electrolytic Decomposition

Decomposition caused by electric current.

Photolytic Decomposition

Decomposition caused by light.

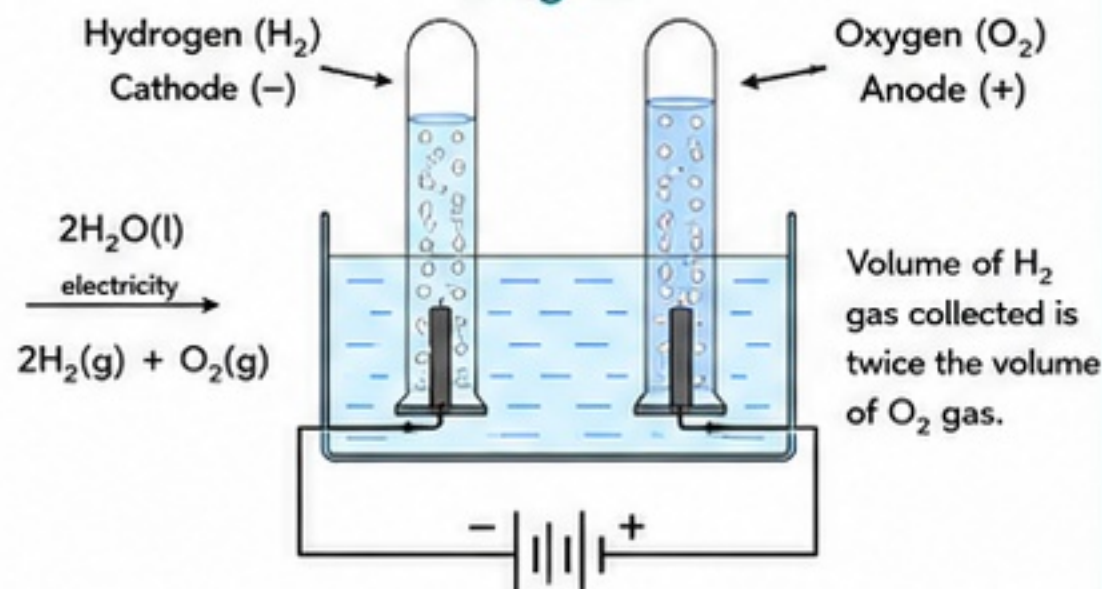
4 Combination vs Decomposition (At a Glance)

Feature	Combination	Decomposition
Number of reactants	Two or more	One
Number of products	One	Two or more
General form	$A + B \rightarrow AB$	$AB \rightarrow A + B$
Energy change	Usually releases energy (exothermic)	Usually requires energy (endothermic)
Mass	Mass of products = Mass of reactants	Mass of products = Mass of reactants
Examples	$2Mg + O_2 \rightarrow 2MgO$	$CaCO_3 \rightarrow CaO + CO_2$

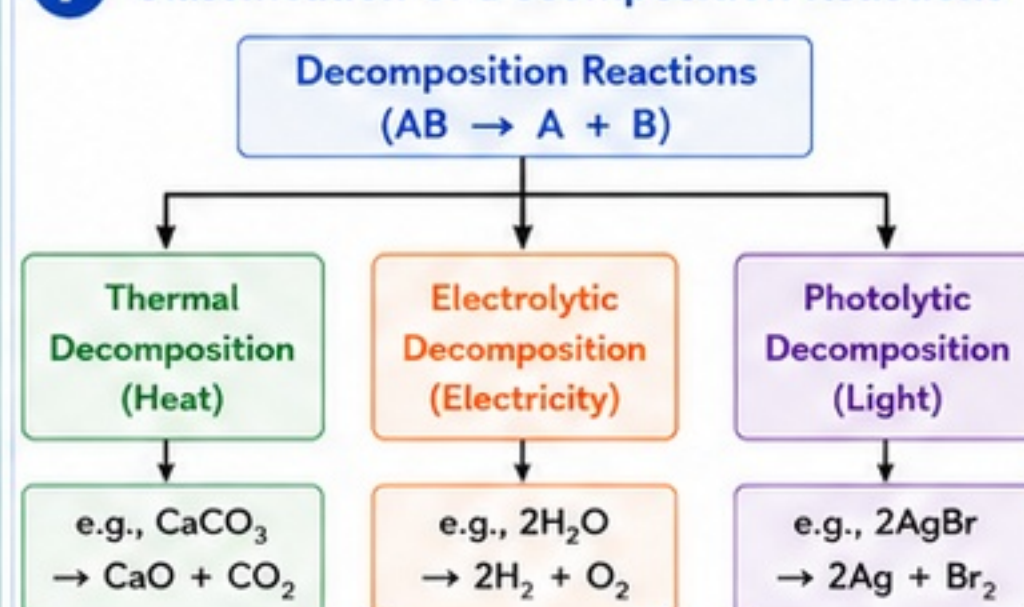
5 Examples of Combination and Decomposition Reactions

Type	Reaction (Balanced Equation)	Products Formed	Observation	Colours / Appearance
Combination	$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$	Magnesium oxide (MgO)	Bright white light; heat released	White ash (MgO)
Combination	$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq)$	Calcium hydroxide solution	Heat evolved; hissing sound	White quicklime to colourless solution
Decomposition	$CaCO_3(s) \xrightarrow{\text{heat}} CaO(s) + CO_2(g)$	Calcium oxide and carbon dioxide	Effervescence; solid residue left	White solid; CO_2 is colourless gas
Decomposition	$2H_2O(l) \xrightarrow{\text{electricity}} 2H_2(g) + O_2(g)$	Hydrogen and oxygen gases	Gas bubbles at both electrodes	Both gases are colourless
Decomposition	$2AgBr(s) \xrightarrow{\text{sunlight}} 2Ag(s) + Br_2(g)$	Silver and bromine	Gradual darkening of tube	AgBr pale yellow/cream; Ag grey/silver; Br_2 brown vapour
Decomposition	$2HgO(s) \xrightarrow{\text{heat}} 2Hg(l) + O_2(g)$	Mercury and oxygen	Red/yellow solid turns to liquid metal	HgO red or yellow solid; Hg silver liquid; O_2 colourless gas
Decomposition	$CuCO_3(s) \xrightarrow{\text{heat}} CuO(s) + CO_2(g)$	Copper(II) oxide and carbon dioxide	Green solid turns black; gas evolved	$CuCO_3$ green; CuO black; CO_2 colourless gas

6 Electrolysis of Water (Decomposition) Diagram



7 Classification of Decomposition Reactions



8 Exam Tips / Common Mistakes / Quick Recall

Exam Tips

- Balance equations carefully.
- Write state symbols (s, l, g, aq).
- Mention energy condition: heat, electricity or light.
- State observations like colour change, gas evolution, heat.

Common Mistakes

- Forgetting to balance equations.
- Missing state symbols.
- Confusing combination and decomposition.
- Not mentioning energy or conditions.

Quick Recall

- ✓ **Combination:**
 $A + B \rightarrow AB$
- ✓ **Decomposition:**
 $AB \rightarrow A + B$
- ✓ **Thermal** → Heat
- ✓ **Electrolytic** → Electricity
- ✓ **Photolytic** → Light
- ✓ Mass of reactants = Mass of products.



Ch 1: Chemical Reactions and Equations

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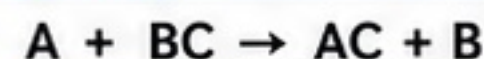


SECTION M – DISPLACEMENT, DOUBLE DISPLACEMENT, PRECIPITATION AND NEUTRALISATION

1 Displacement Reaction

- A more reactive element displaces a less reactive element from its compound in solution.

General Form:



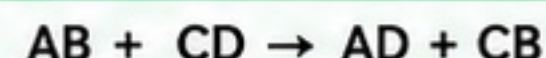
Reactivity Series (most to least reactive):



2 Double Displacement Reaction

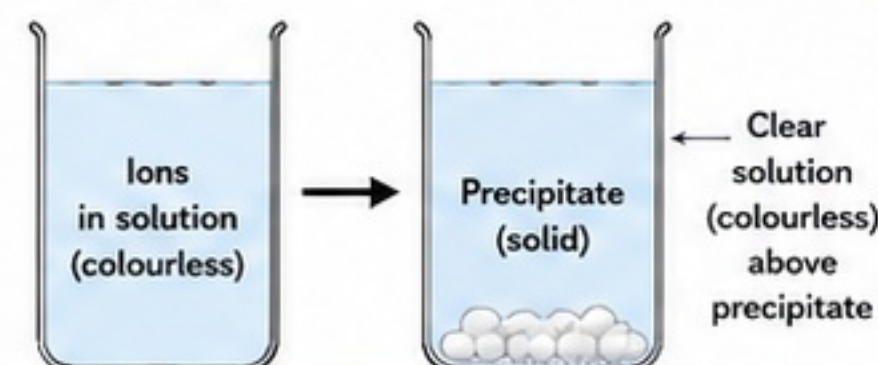
- Ions in two compounds exchange partners to form two new compounds in solution (or as a precipitate, gas or water).

General Form:



3 Precipitation Reaction

- A double displacement reaction in which an insoluble solid (precipitate) forms in solution.



4 Neutralisation Reaction

- A reaction between an acid and a base to form a salt and water.

General Form:



5 Examples of Displacement, Double Displacement, Precipitation and Neutralisation Reactions

Type	Balanced Equation	Observations	
		Observation	Colour of Every Important Compound / Product
Displacement	$\text{Fe(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{FeSO}_4\text{(aq)} + \text{Cu(s)}$	Reddish-brown solid (Cu) forms on iron.	Fe: grey; $\text{CuSO}_4\text{(aq)}$: blue; $\text{FeSO}_4\text{(aq)}$: pale green; Cu(s): reddish-brown deposit.
Displacement	$\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$	Reddish-brown solid (Cu) forms on zinc.	Zn: silvery grey; $\text{CuSO}_4\text{(aq)}$: blue; $\text{ZnSO}_4\text{(aq)}$: colourless; Cu(s): reddish-brown deposit.
Precipitation	$\text{AgNO}_3\text{(aq)} + \text{NaCl(aq)} \rightarrow \text{AgCl(s)} + \text{NaNO}_3\text{(aq)}$	White precipitate forms.	$\text{AgNO}_3\text{(aq)}$: colourless; NaCl(aq) : colourless; AgCl(s) : white; $\text{NaNO}_3\text{(aq)}$: colourless.
Precipitation	$\text{BaCl}_2\text{(aq)} + \text{Na}_2\text{SO}_4\text{(aq)} \rightarrow \text{BaSO}_4\text{(s)} + 2\text{NaCl(aq)}$	White precipitate forms.	$\text{BaCl}_2\text{(aq)}$: colourless; $\text{Na}_2\text{SO}_4\text{(aq)}$: colourless; $\text{BaSO}_4\text{(s)}$: white; NaCl(aq) : colourless.
Precipitation	$\text{Pb(NO}_3)_2\text{(aq)} + 2\text{KI(aq)} \rightarrow \text{PbI}_2\text{(s)} + 2\text{KNO}_3\text{(aq)}$	Bright yellow precipitate forms.	$\text{Pb(NO}_3)_2\text{(aq)}$: colourless; KI(aq) : colourless; $\text{PbI}_2\text{(s)}$: bright yellow; $\text{KNO}_3\text{(aq)}$: colourless.
Double Displacement (Precipitation)	$\text{CuSO}_4\text{(aq)} + 2\text{NaOH(aq)} \rightarrow \text{Cu(OH)}_2\text{(s)} + \text{Na}_2\text{SO}_4\text{(aq)}$	Light blue precipitate forms.	$\text{CuSO}_4\text{(aq)}$: blue; NaOH(aq) : colourless; $\text{Cu(OH)}_2\text{(s)}$: light blue; $\text{Na}_2\text{SO}_4\text{(aq)}$: colourless.
Neutralisation	$\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$	No visible change. Mixture remains clear.	HCl(aq) : colourless; NaOH(aq) : colourless; NaCl(aq) : colourless; $\text{H}_2\text{O(l)}$: colourless.
Neutralisation	$\text{H}_2\text{SO}_4\text{(aq)} + 2\text{NaOH(aq)} \rightarrow \text{Na}_2\text{SO}_4\text{(aq)} + 2\text{H}_2\text{O(l)}$	No visible change. Mixture remains clear.	$\text{H}_2\text{SO}_4\text{(aq)}$: colourless; NaOH(aq) : colourless; $\text{Na}_2\text{SO}_4\text{(aq)}$: colourless; $\text{H}_2\text{O(l)}$: colourless.

6 Net Ionic Equations for Common Precipitates

Precipitate	Net Ionic Equation	Observation & Colour
AgCl (white)	$\text{Ag}^+\text{(aq)} + \text{Cl}^-\text{(aq)} \rightarrow \text{AgCl(s)}$	White precipitate.
BaSO ₄ (white)	$\text{Ba}^{2+}\text{(aq)} + \text{SO}_4^{2-}\text{(aq)} \rightarrow \text{BaSO}_4\text{(s)}$	White precipitate.
PbI ₂ (bright yellow)	$\text{Pb}^{2+}\text{(aq)} + 2\text{I}^-\text{(aq)} \rightarrow \text{PbI}_2\text{(s)}$	Bright yellow precipitate.
Cu(OH) ₂ (light blue)	$\text{Cu}^{2+}\text{(aq)} + 2\text{OH}^-\text{(aq)} \rightarrow \text{Cu(OH)}_2\text{(s)}$	Light blue precipitate.

7 How to Identify the Type of Reaction

Reaction Type	How to Identify
Displacement	A single element replaces another element in a compound ($A + BC \rightarrow AC + B$).
Double Displacement	Two compounds exchange ions/partners ($AB + CD \rightarrow AD + CB$).
Precipitation	A solid precipitate (insoluble) forms in solution.
Neutralisation	An acid reacts with a base to form a salt and water.

8 Real-Life Links

- Displacement:** Rusting of iron, thermite reaction (Al displaces other metals).
- Precipitation:** Water purification (lime softening), removal of heavy metals from waste water.
- Neutralisation:** Antacids neutralise stomach acid; soil treatment with lime.

8 Exam Tips

- Look for clues: formation of a solid, gas or water.
- Write the balanced molecular equation first.
- Cross out spectator ions to write the net ionic equation.
- Always state the colour of every important compound or product.
- Practice reactivity series to predict displacement.



Ch 1: Chemical Reactions and Equations

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SECTION N – BALANCING CHEMICAL EQUATIONS MASTERCLASS

1 Why Balancing is Necessary – Law of Conservation of Mass

- Matter is neither created nor destroyed in a chemical reaction.
- The total number of atoms of each element must be the same on both sides of the equation.
- Balancing ensures equations are scientifically correct and results are meaningful.
- It helps predict quantities of reactants and products accurately.

2 Step-by-Step Balancing Rules

- Write the correct skeletal equation (with formulas of all species).
- Count the number of atoms of each element on both sides.
- Balance elements that appear in one compound on each side first.
- Balance polyatomic ions as a unit (e.g., SO_4 , NO_3 , OH).
- Balance H and O last.
- Use the smallest whole-number coefficients.
- Recount all atoms to check both sides are equal.

3 State Symbols and Special Symbols

Symbol	Meaning / Use
(s)	Solid
(l)	Liquid
(g)	Gas
(aq)	Aqueous (dissolved in water)
Δ	Heat supplied
$h\nu$	Light (photochemical reaction)
Pt, Ni, MnO_2 etc.	Catalyst (written above arrow)
$\downarrow$	Precipitate forms (insoluble solid)
$\uparrow$	Gas evolved

4 Worked Example 1: $\text{Cu} + \text{O}_2 \rightarrow \text{CuO}$

Step 1 (Skeletal): $\text{Cu} + \text{O}_2 \rightarrow \text{CuO}$

Step 2 (Count atoms): LHS – Cu: 1, O: 2
RHS – Cu: 1, O: 1

Step 3: Balance O first.
 $\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$

Step 4 (Recount): LHS – Cu: 1, O: 2
RHS – Cu: 2, O: 2

Step 5: Balance Cu.
 $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$ ✓ **Balanced**

Observation / Colour: Copper (reddish-brown) reacts with oxygen to form black copper(II) oxide.

5 Worked Example 2: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$

Step 1 (Skeletal): $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$

Step 2 (Count atoms):

LHS – Fe: 1, Cu: 1, S: 1, O: 4
RHS – Fe: 1, Cu: 1, S: 1, O: 4

Step 3: All atoms already equal.

Step 4 (Balanced Equation):

$\text{Fe} + \text{CuSO}_4(\text{aq}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$

Observation / Colour: Blue CuSO_4 solution turns pale green (FeSO_4) and reddish-brown copper is deposited.

6 Worked Example 3: $\text{Al} + \text{HCl} \rightarrow \text{AlCl}_3 + \text{H}_2$

Step 1 (Skeletal): $\text{Al} + \text{HCl} \rightarrow \text{AlCl}_3 + \text{H}_2$

Step 2 (Count atoms):

LHS – Al: 1, H: 1, Cl: 1 RHS – Al: 1, H: 2, Cl: 3

Step 3: Balance Cl first $\rightarrow$ put 3 before HCl.
 $\text{Al} + 3\text{HCl} \rightarrow \text{AlCl}_3 + \text{H}_2$

Step 4 (Recount): LHS – Al: 1, H: 3, Cl: 3
RHS – Al: 1, H: 2, Cl: 3

Step 5: Balance H $\rightarrow$ put 3/2 before H_2 ; then make whole numbers by multiplying all by 2.
 $2\text{Al} + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2$ ✓ **Balanced**

Observation / Colour: Aluminium (silvery) reacts with acid; hydrogen gas evolved with 'pop' sound.

7 Worked Example 4: $\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$

Step 1 (Skeletal): $\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$

Step 2 (Count atoms):

LHS – Pb: 1, N: 2, O: 6, K: 1, I: 1
RHS – Pb: 1, I: 2, K: 1, N: 1, O: 3

Step 3: Balance NO_3 as a unit $\rightarrow$ put 2 before KNO_3 .
 $\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$

Step 4 (Recount): LHS – Pb: 1, N: 2, O: 6, K: 1, I: 1
RHS – Pb: 1, N: 2, O: 6, K: 2, I: 2

Step 5: Balance K and I $\rightarrow$ put 2 before KI.
 $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$ ✓ **Balanced**

Observation / Colour: Bright yellow PbI_2 precipitate forms.

8

Practice Table – Word, Skeletal and Balanced Equations with Observations

No.	Word Equation	Skeletal Equation	Balanced Equation (with states)	Observation / Colour Change
1	Magnesium + Oxygen $\rightarrow$ Magnesium oxide	$\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$	$2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{MgO}(\text{s})$	Silvery magnesium burns with bright white light; white ash.
2	Zinc + Hydrochloric acid $\rightarrow$ Zinc chloride + Hydrogen	$\text{Zn} + \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$	$\text{Zn}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{ZnCl}_2(\text{aq}) + \text{H}_2(\text{g})$	Bubbles of hydrogen; zinc dissolves; solution remains colourless.
3	Sodium + Water $\rightarrow$ Sodium hydroxide + Hydrogen	$\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2$	$2\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq}) + \text{H}_2(\text{g})$	Fizzing; hydrogen evolved; solution becomes alkaline.
4	Iron + Sulfur $\rightarrow$ Iron(II) sulfide	$\text{Fe} + \text{S} \rightarrow \text{FeS}$	$\text{Fe}(\text{s}) + \text{S}(\text{s}) \rightarrow \text{FeS}(\text{s})$	Grey/black solid (iron(II) sulfide) forms.
5	Silver nitrate + Sodium chloride $\rightarrow$ Silver chloride + Sodium nitrate	$\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$	$\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s})\downarrow + \text{NaNO}_3(\text{aq})$	White AgCl precipitate forms.
6	Hydrogen peroxide $\rightarrow$ Water + Oxygen	$\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$	$2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$	Bubbles of oxygen; light glowing splint rekindles.

9

Common Mistakes to Avoid

- ✗ Changing subscripts instead of coefficients. (Wrong: $\text{H}_2\text{O} \rightarrow \text{H}_4\text{O}$; Correct: $2\text{H}_2\text{O}$)
- ✗ Not recounting atoms after balancing.
- ✗ Forgetting polyatomic ions – splitting them. (Wrong: $\text{H}_2\text{SO}_4 \rightarrow \text{H}_2 + \text{S} + \text{O}_4$)
- ✗ Omitting state symbols or special symbols.
- ✗ Using fractional coefficients – always make whole numbers.

10

Exam Tips – Score Full Marks

- ✓ Always write the correct formulas first.
- ✓ Balance systematically – follow the rules.
- ✓ Show all steps in long equations.
- ✓ Write state symbols (s), (l), (g), (aq) wherever required.
- ✓ Mention conditions above arrow (Δ , $h\nu$, catalyst).
- ✓ State observations clearly (colour, precipitate, gas, heat/light).
- ✓ Recheck atom count at the end – this is exam gold!



Quick Check – Did You Remember?

- Balanced equations obey the law of conservation of mass.
- Coefficients change, subscripts never change.
- Use smallest whole-number coefficients.
- Balance polyatomic ions as a unit.
- Write states, conditions and observations for full marks.
- Practice regularly – speed and accuracy come with practice!



Ch 1: Chemical Reactions and Equations

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SECTION 0 – OXIDATION, REDUCTION AND REDOX REACTIONS (DEEPER VIEW)

1 Oxidation – Three Ways to Define It

- Gain of oxygen**
A substance is oxidised when it combines with oxygen.
- Loss of hydrogen**
A substance is oxidised when it loses hydrogen.
- Loss of electrons**
A substance is oxidised when it loses one or more electrons.

Mnemonic: Oxidation = OIL
(Oxygen gain, Hydrogen loss, Electron loss)

2 Reduction – Three Ways to Define It

- Loss of oxygen**
A substance is reduced when it loses oxygen.
- Gain of hydrogen**
A substance is reduced when it gains hydrogen.
- Gain of electrons**
A substance is reduced when it gains one or more electrons.

Mnemonic: Reduction = LGE
(Loss of oxygen, Gain of hydrogen, Gain of electrons)

3 Oxidising Agent and Reducing Agent

- Oxidising agent (oxidant)**
Substance that causes oxidation by gaining electrons.
It is reduced in the process.
- Reducing agent (reductant)**
Substance that causes reduction by losing electrons.
It is oxidised in the process.

Oxidising agent : gains e^-
causes oxidation of another

Reducing agent : loses e^-
causes reduction of another

4 Why Oxidation and Reduction Always Occur Together

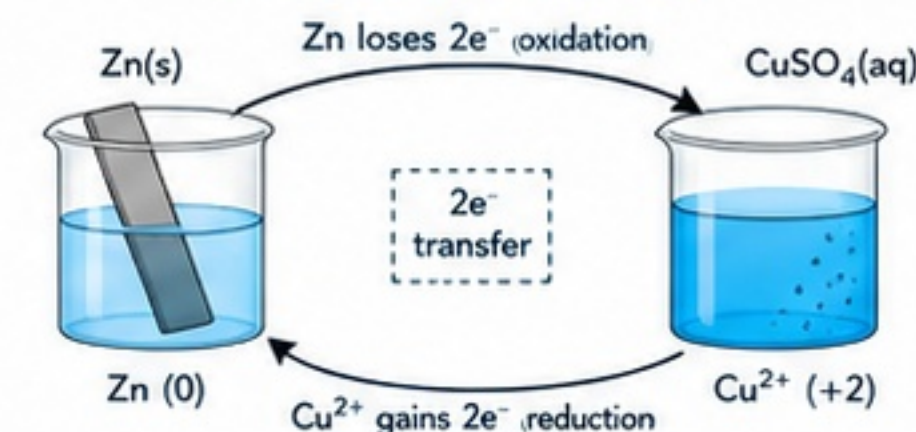
- Electrons lost by one species must be gained by another.
- One species is oxidised; the other is reduced.
- Total charge and mass are conserved.
- Therefore, oxidation and reduction are complementary and occur together in a redox reaction.

Key idea: OIL–RIG
(One Is Lost; the other Is Gained)

5 Redox Reactions Table

Reaction (Balanced Equation)	What is Oxidised?	What is Reduced?	Oxidising Agent (accepts e^-)	Reducing Agent (donates e^-)	Observation / Colour Change
$2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$	Cu ($0 \rightarrow +2$)	O_2 ($0 \rightarrow -2$)	O_2	Cu	Cu reddish-brown to CuO black
$\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$	H_2 ($0 \rightarrow +1$)	Cu ($+2 \rightarrow 0$)	CuO	H_2	CuO black to Cu reddish-brown, H_2 colourless
$\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$	Zn ($0 \rightarrow +2$)	Cu ($+2 \rightarrow 0$)	Cu^{2+} (in CuSO_4)	Zn	CuSO_4 blue to ZnSO_4 colourless, Cu reddish-brown deposited
$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$	Fe ($0 \rightarrow +2$)	Cu ($+2 \rightarrow 0$)	Cu^{2+} (in CuSO_4)	Fe	CuSO_4 blue to FeSO_4 pale green, Cu reddish-brown deposited
$2\text{Fe} + 3\text{Cl}_2 \rightarrow 2\text{FeCl}_3$	Fe ($0 \rightarrow +3$)	Cl_2 ($0 \rightarrow -1$)	Cl_2	Fe	Cl_2 greenish-yellow gas, FeCl_3 brownish/yellow-brown fumes or solution
$4\text{Fe} + 3\text{O}_2 + x\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$	Fe ($0 \rightarrow +3$)	O_2 ($0 \rightarrow -2$)	O_2	Fe	Rust (hydrated iron(III) oxide) reddish-brown

6 Electron-Transfer Diagram (Zn / CuSO_4 Reaction)



Overall: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
Zn is oxidised; Cu^{2+} is reduced.

7 Oxidation Number (O.N.) Memory Cues

- Elements in their standard state = 0 (e.g., O_2 , H_2 , Fe, Cu).
- Monatomic ion: O.N. = charge (e.g., $\text{Na}^+ = +1$, $\text{Cl}^- = -1$).
- O in compounds: usually -2 (except peroxides -1 , OF_2 $+2$).
- H in compounds: usually $+1$ (except metal hydrides -1).
- Sum of O.N.s in neutral compound = 0
Sum of O.N.s in polyatomic ion = charge.

8 Redox in Daily Life

- Respiration:** Glucose is oxidised to CO_2 (release of energy).
- Bleaching:** $\text{Ca}(\text{ClO})_2$ oxidises stains; colourless products formed.
- Corrosion:** Iron rusts (oxidation by O_2 and moisture) forming hydrated iron(III) oxide.
- Disinfection:** Cl_2 , ozone, H_2O_2 oxidise and kill harmful microorganisms.

9 Key Colour Memory List

- Cu (metal): reddish-brown
- CuO: black
- $\text{CuSO}_4(\text{aq})$: blue
- $\text{ZnSO}_4(\text{aq})$: colourless
- $\text{FeSO}_4(\text{aq})$: pale green
- FeCl_3 : brownish / yellow-brown
- Cl_2 : greenish-yellow gas
- Rust ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$): reddish-brown
- H_2 : colourless gas

10 Exam Tips and Common Errors

- Always balance the equation before analysing redox.
- Identify oxidation and reduction using O.N. changes (most reliable).
- Do not confuse gaining oxygen with gaining charge.
- Remember: Oxidation = loss of e^- ; Reduction = gain of e^- .
- Check colours and observations carefully.
- Write half-reactions if asked.



Ch 1: Chemical Reactions and Equations

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SECTION P – EXOTHERMIC AND ENDOTHERMIC REACTIONS

1 Energy Change Idea

- Chemical reactions involve breaking old bonds and forming new bonds.
- Energy is needed to break bonds (endothermic step).
- Energy is released when new bonds form (exothermic step).
- If more energy is released than absorbed → net energy released (exothermic).
- If more energy is absorbed than released → net energy absorbed (endothermic).

2 Exothermic Reaction (Heat Released)

- A reaction that releases energy to the surroundings, usually as heat.
- Surroundings become warmer.
- Temperature increases.
- Often gives off light (flame/glow).
- Typically products have lower chemical potential energy than reactants.



Heat flows out: system → surroundings

3 Endothermic Reaction (Heat Absorbed)

- A reaction that absorbs energy from the surroundings, usually as heat.
- Surroundings become cooler.
- Temperature decreases.
- No light is produced.
- Typically products have higher chemical potential energy than reactants.



Heat flows in: surroundings → system

4 Bond Breaking vs Bond Making – Why Energy Changes Happen

Breaking bonds:

- Requires energy input (endothermic).
- Old bonds between reactant particles are broken.

Making bonds:

- Releases energy (exothermic).
- New bonds between product particles are formed.

Net energy change =

Energy to break bonds – Energy released in making new bonds

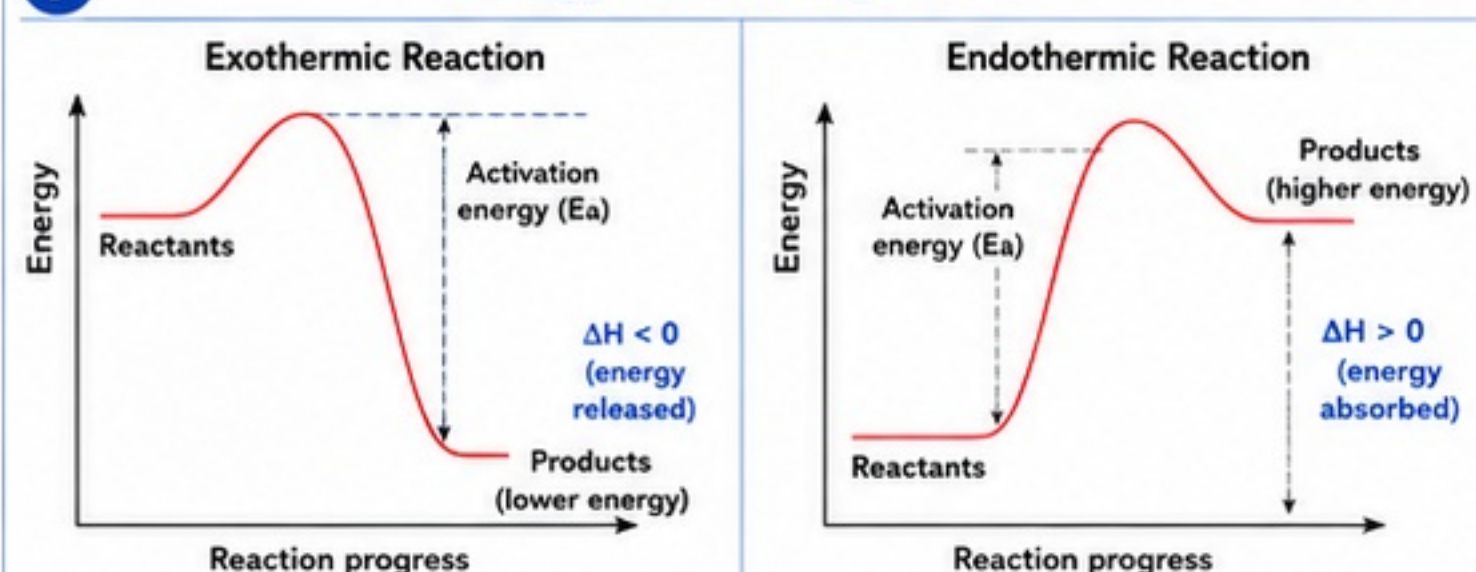
5 Exothermic vs Endothermic Reactions – Comparison

Feature	Exothermic	Endothermic
Heat flow	Heat released to surroundings	Heat absorbed from surroundings
Temperature	Surroundings warm up	Surroundings cool down
Energy (ΔH)	ΔH is negative (–)	ΔH is positive (+)
Products vs Reactants (energy level)	Products have lower energy than reactants	Products have higher energy than reactants
Examples	Combustion, neutralisation, respiration	Photosynthesis, thermal decomposition, dissolving ammonium nitrate
Common signs	Heat, light, warm surface	Cool surface, needs external heating

6 Examples of Exothermic and Endothermic Reactions

Type	Reaction (Equation)	Observation	Colours / Appearance
Exo	$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$	Dazzling white flame; intense heat	White ash/solid MgO formed
Exo	$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$	Blue flame; heat released	CO_2 (colourless gas), H_2O (colourless steam)
Exo	$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$	Heat evolved; mixture becomes hot	White quicklime + water → colourless solution
Exo	$\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$	Neutralisation; heat evolved	Both solutions colourless; solution remains colourless
Endo	$\text{CaCO}_3 \xrightarrow{\text{heat}} \text{CaO} + \text{CO}_2$	Requires heating; no flame; test tube cools on its own	White solid $\text{CaCO}_3 \rightarrow \text{CO}_2$ (colourless gas) + white CaO
Endo	$\text{NH}_4\text{NO}_3 (\text{s})$ dissolving in H_2O	Solution becomes cold (cooling effect)	Solid white salt → colourless solution
Endo	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \xrightarrow{\text{heat}} \text{CuSO}_4 + 5\text{H}_2\text{O}$	Requires heating; water vapour given off	Blue crystals → white anhydrous powder (CuSO_4)
Endo	$6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{light}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$	Needs sunlight; no heat released	Green leaves; O_2 colourless gas

7 Energy Profile Diagrams



8 Daily-Life Applications

- Exothermic:**
 - Cooking (combustion of gas)
 - Hand warmers
 - Neutralisation in digestion
 - Rusting (slow exothermic)
- Endothermic:**
 - Instant cold packs (NH_4NO_3)
 - Photosynthesis in plants
 - Thermal decomposition in industries (e.g., limestone → lime + CO_2)

9 Quick Recall

- Heat released → exothermic.
- Heat absorbed → endothermic.
- Exothermic: ΔH negative, surroundings get warm.
- Endothermic: ΔH positive, surroundings get cool.
- More energy released than absorbed → exothermic.
- More energy absorbed than released → endothermic.
- Energy diagrams show relative energy of reactants and products.

10 Exam Tips

- Write "heat released" or "heat evolved" for exothermic.
- Write "heat absorbed" or "heat required" for endothermic.
- Always mention temperature change (warm or cool).
- Include observation with colour/appearance where relevant.
- Draw and label energy profile diagrams neatly.
- Use ΔH sign correctly (– for exo, + for endo).
- Link bond breaking/making to energy change clearly in answers.



Ch 1: Chemical Reactions and Equations

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SECTION Q – CORROSION (RUSTING AND ITS PREVENTION)

1 Definition of Corrosion

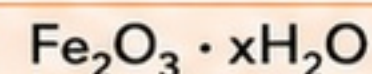
- Corrosion is the gradual deterioration of metals due to chemical reaction with their environment.
- It is a natural process that returns metals to more stable compounds like oxides, hydroxides or carbonates.
- Rusting is a type of corrosion that occurs in iron.

2 Why Corrosion Happens

- Metals are not completely stable; they tend to return to stable forms (oxides, carbonates, sulphides, etc.).
- When exposed to air, moisture, acids, salts or other chemicals, metals react slowly.
- This reaction forms a layer of corrosion products on the metal surface.

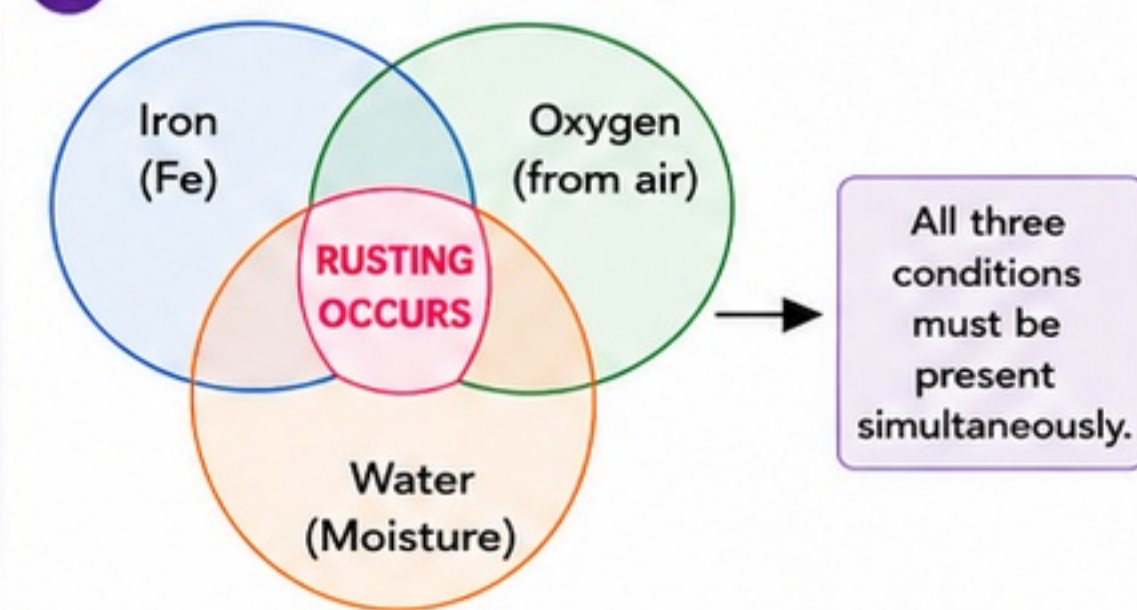
3 Rusting of Iron Explained

- Rusting is the corrosion of iron in presence of moisture and air (oxygen).
- The main product is hydrated ferric oxide:



- Rust is reddish-brown in colour, loose and flaky.

4 Conditions Necessary for Rusting



5 Common Corrosion Examples

Metal	Corrosion Product	Colour / Appearance
Iron (Fe)	$\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (Rust)	Reddish-brown, flaky
Copper (Cu)	Basic copper carbonate / copper compounds	Green (patina)
Silver (Ag)	Ag_2S (Silver sulphide)	Black tarnish
Aluminium (Al)	Al_2O_3 (Aluminium oxide)	Thin, dull grey protective layer
Zinc (Zn)	ZnO / ZnCO_3	Whitish protective layer
Chromium (Cr)	Chromium oxide	Shiny, protective oxide layer

6 Disadvantages of Corrosion

- Wastage of valuable metals and materials.
- Weakening of structures and machines.
- Reduced strength and safety of bridges, buildings, vehicles, tools and equipment.
- Leakage in pipelines, tanks and boilers.
- Increased maintenance and repair cost.
- Loss of efficiency and shorter life of products.

7 Methods of Prevention of Corrosion

Method	How it Works (Simple Cross-Section)	Coating / Protection	Colour / Appearance
Painting	 Metal (Iron)	Paint layer (organic)	Any colour (usually bright)
Oiling / Greasing	 Metal (Iron)	Oil / Grease film	Yellowish / Brownish (transparent)
Galvanisation	 Metal (Iron)	Zinc coating (Zn)	Silvery-grey
Alloying	 Metal (Iron)	Alloy layer (e.g., stainless steel)	Silvery, metallic
Electroplating (Example: Nickel)	 Metal (Iron)	Nickel coating (Ni)	Silvery, shiny

8 Rusting Experiment (Observation Table)

Setup	Conditions	Observation After Few Days	Result
Nail in dry air	Iron + Air (no water)	No rust	No rusting
Nail in boiled water (cooled)	Iron + Water (no air)	No rust	No rusting
Nail in tap water	Iron + Air + Water	Reddish-brown rust forms	Rusting occurs
Nail in salt water	Iron + Air + Water + Salt (NaCl)	Heavy reddish-brown rust forms quickly	Rusting occurs faster

9 Daily-Life Applications and Maintenance

- Corrosion control increases the life of tools, vehicles, buildings and machines.
- Regular cleaning, painting and lubrication help prevent corrosion.
- Use of galvanised sheets for roofs, stainless steel for kitchen utensils, and coated pipes for water supply.
- Helps save money, ensures safety and protects the environment.

10 Exam Tips / Quick Recall

- Corrosion = deterioration of metals by chemical reaction with environment.
- Rusting needs: iron + oxygen + water.
- Rust = $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (reddish-brown).
- Conditions: air, water and iron must all be present.
- Prevention: painting, oiling, galvanisation, alloying, electroplating.
- Salt water causes faster rusting.
- Remember examples and colours of corrosion products.



Ch 1: Chemical Reactions and Equations

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SECTION R – RANCIDITY AND OXIDATION OF FATS AND OILS

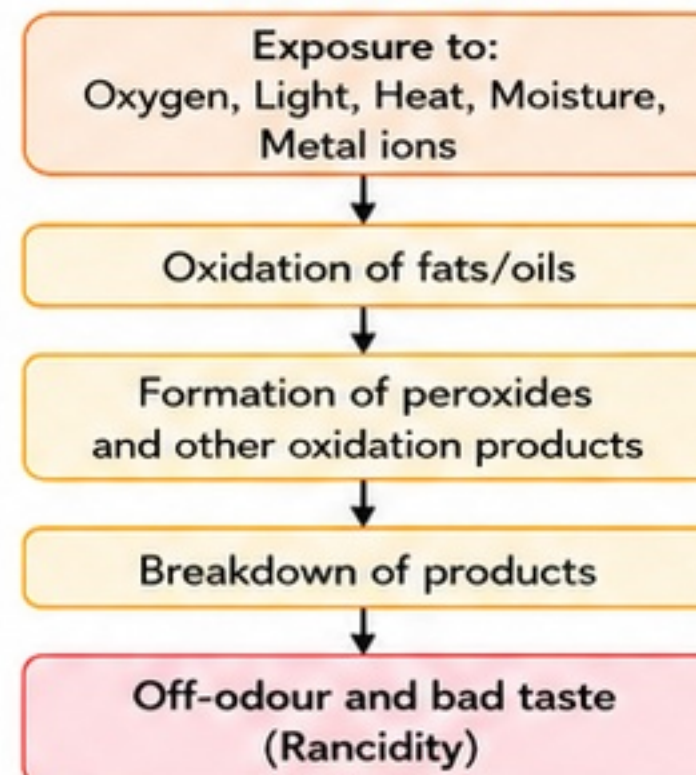
1 What is Rancidity?

- Rancidity is the oxidation of fats and oils.
- It causes unpleasant smell (off-odour) and bad taste.
- It leads to spoilage and reduces the quality and shelf life of food.
- It is a chemical change – fats react with oxygen to form new products.
- Fresh fats/oils are safe and pleasant; rancid fats/oils are unfit to eat.

2 Why Do Fats and Oils Become Rancid?

- Fats and oils react with oxygen in air – a slow oxidation process.
- Unsaturated fats have $C=C$ double bonds $\rightarrow$ more reactive $\rightarrow$ more prone to rancidity.
- Heat, light, moisture and metal ions (like Fe, Cu) speed up oxidation.
- The formed peroxides and other products break down and cause off-flavour and off-odour.

3 How Rancidity Happens (Flow Diagram)



4 Signs / Observations of Rancidity

- **Change in colour:**
Fresh oil is pale yellow/clear.
Rancid oil becomes yellowish-brown.
- **Off-odour:** unpleasant, stale or musty smell.
- **Bad taste:** bitter or stale taste.
- Foam or fizz may appear in some cases.
- Loss of nutritive value.
- Food becomes unfit for consumption.

Appearance / Colour Guide

Fresh oil/fat	Pale yellow to colourless (clear)
Rancid oil/fat	Yellowish-brown, darkened
Nitrogen gas (used for protection)	Colourless
Oxygen gas (cause of oxidation)	Colourless

5 Methods to Prevent Rancidity

- Store in airtight containers to limit oxygen.
- Refrigerate or store in a cool place.
- Use antioxidants (e.g., BHA, BHT, Vitamin E).
- Flush with nitrogen gas (inert, colourless) to displace oxygen.
- Keep away from heat, light and moisture.
- Use clean, dry spoons and containers.
- Buy in small quantities; use within the best-before date.

6 Examples in Daily Life

- Cooking oils (sunflower, groundnut, soybean, etc.) turn rancid on long storage.
- Potato chips become stale and smell bad.
- Nuts and seeds develop off-flavour.
- Butter and ghee develop rancid smell.
- Bakery products (biscuits, cakes, namkeen) lose taste and aroma.

7 Corrosion vs Rancidity

Feature	Corrosion	Rancidity
What reacts?	Metals	Fats and oils
Reactant	Oxygen + moisture	Oxygen (air)
Type	Surface reaction	Bulk molecular reaction
Products	Metal oxides / hydroxides	Peroxides and other products
Appearance	Rust / dull surface	Yellowish-brown colour
Effect	Damages metals	Bad smell, bad taste, spoilage
Prevention	Paint, galvanization, oiling, coatings	Airtight storage, refrigeration, antioxidants, nitrogen flushing

8 Role of Oxidation in Food Spoilage & Why Chemical Reactions Matter

- Oxidation leads to spoilage of fats, oils and many foods.
- It reduces nutrition, taste, aroma and shelf life.
- Chemical reactions happen in daily life – in cooking, storage, digestion, rusting, cleaning, medicines, and industries.
- Understanding reactions helps us store food safely, prevent waste and stay healthy.

9 One-Line Recall

- **Rancidity:** oxidation of fats and oils causing bad smell and taste.
- **Cause:** reaction with oxygen (air).
- **Products:** peroxides and other oxidation products.
- **Appearance:** fresh oil pale yellow/clear; rancid oil yellowish-brown.
- **Prevention:** airtight, cool, dark, antioxidants, nitrogen gas.
- **Nitrogen:** colourless, inert (protects).
- **Oxygen:** colourless, reactive (causes).

10 Exam Tips & High-Yield Points

- Define rancidity and mention it is due to oxidation of fats and oils.
- Write the flow: exposure $\rightarrow$ oxidation $\rightarrow$ peroxides $\rightarrow$ off-odour.
- Write signs: change in colour (pale yellow $\rightarrow$ yellowish-brown), off-odour, bad taste.
- List prevention methods with examples (antioxidants, nitrogen flushing, refrigeration).
- Compare with corrosion in table form (reactant, products, effect, prevention).
- Write daily-life examples: oils, chips, nuts, butter, bakery products.
- Mention appearances/colours: fresh oil pale yellow/clear; rancid oil yellowish-brown.
- Mention gases clearly: Nitrogen gas – colourless, inert; Oxygen gas – colourless, causes oxidation.
- Use keywords: oxidation, unsaturated fats, oxygen, peroxides, off-odour, spoilage.



Ch 1: Chemical Reactions and Equations

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SECTION S – LAB TESTS, STATE SYMBOLS, OBSERVATIONS AND CONDITIONS

1 State Symbols and What They Mean

- (s) **solid**
The substance is in solid state.
- (l) **liquid**
The substance is in liquid state.
- (g) **gas**
The substance is in gaseous state.
- (aq) **aqueous**
The substance is dissolved in water.

Example: NaCl(s) , $\text{H}_2\text{O(l)}$, $\text{CO}_2\text{(g)}$, $\text{CuSO}_4\text{(aq)}$

2 Special Symbols and Conditions

Δ	Heat
$h\nu$	Light
⚡	Electricity
cat.	Catalyst
$\downarrow$	Precipitate forms
$\uparrow$	Gas evolved

Use these above or below the arrow in chemical equations.

3 Common Gas Tests and Observations

Gas	Test / Observation
H_2	'Pop' sound with burning splint.
O_2	Rekindles a glowing splint.
CO_2	Turns lime water milky.
Cl_2	Greenish-yellow gas; bleaches moist litmus paper.
HCl	Pungent smell; forms white fumes with NH_3 .

4 Common Precipitate Colours

Precipitate	Colour
AgCl	White
BaSO_4	White
PbI_2	Bright yellow
Cu(OH)_2	Light blue
CaCO_3	White

Note: Some precipitates may darken on standing or on exposure to light.

5 Common Solution Colours

Solution	Colour
CuSO_4 solution	Blue
FeSO_4 solution	Pale green
ZnSO_4 solution	Colourless
NaCl solution	Colourless
HCl solution	Colourless
NaOH solution	Colourless
AgNO_3 solution	Colourless

Note: Dilute solutions unless stated otherwise.

6 Observation Writing Guide Write Complete Sentences!

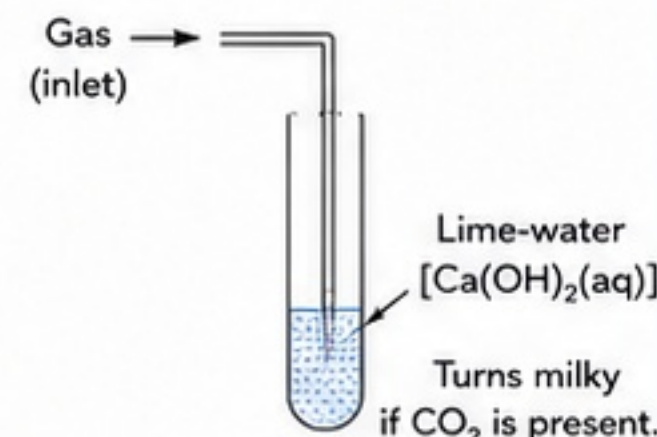
- Start with the substance or action.
- State the observation (what you see, smell, hear or feel).
- Mention colour, change, amount and any other detail.
- Use precise words and standard terms.
- End the sentence properly.

Examples:

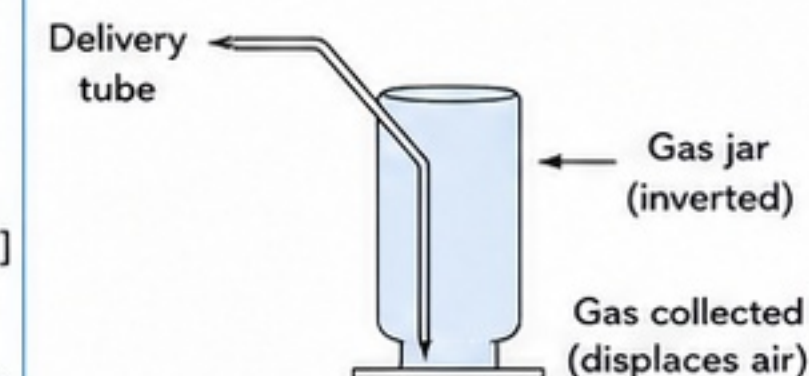
- ✓ A white precipitate of AgCl forms.
- ✓ The blue CuSO_4 solution turns pale green.
- ✓ A 'pop' sound is heard when a burning splint is brought near the gas.
- ✓ The gas turns lime water milky.
- ✓ The colourless solution gives a white precipitate with NaOH solution.

7 Mini Lab Setups / Diagrams

Lime-Water Test for CO_2



Gas Collection by Downward Displacement of Air



Method for gases denser than air (e.g., CO_2 , Cl_2).

8 Common Errors to Avoid

- ✗ Writing incomplete observations.
- ✗ Not mentioning colour or change.
- ✗ Not using correct state symbols.
- ✗ Confusing gas tests (e.g., O_2 vs H_2).
- ✗ Ignoring conditions (heat, catalyst, etc.).
- ✗ Wrong spelling of chemical names and observations.
- ✗ Forgetting to balance equations.

9 Quick Revision – High-Yield List

- ✓ Know state symbols and their meanings.
- ✓ Remember test for each common gas.
- ✓ Learn common precipitate colours.
- ✓ Recall common solution colours.
- ✓ Use correct special symbols and conditions.
- ✓ Write complete, precise observation sentences.
- ✓ Practice diagrams for gas collection and tests.
- ✓ Balance all chemical equations.

10 Exam Tips

- ★ In practical questions, write observations in full sentences.
- ★ Write balanced equations with correct state symbols.
- ★ Draw neat, labelled diagrams wherever asked.
- ★ Always mention colour change, precipitate, gas, smell, heat/light.
- ★ If a gas is given, first write the test, then observation.
- ★ In multi-step questions, number your answers and keep them in order.
- ★ Check your work: units, symbols, spelling and balancing.
- ★ Present cleanly – neat handwriting earns marks!



Ch 1: Chemical Reactions and Equations

Part 2 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus



SECTION T – FINAL MASTER REVISION AND CENTUM SHEET

1 Complete Chapter at a Glance

- Chemical reactions convert reactants into products through rearrangement of atoms.
- Equations represent reactions using symbols and formulae.
- Observations (colour, gas, precipitate, change in temperature, smell, etc.) confirm reactions.
- Balanced equations follow the law of conservation of mass.
- Reactions are classified into different types based on patterns.
- Everyday life and industries depend on chemical reactions.

2 Most Important Definitions

- Chemical reaction:** Process in which one or more substances are converted into new substances.
- Reactant:** Substance(s) present at the start.
- Product:** Substance(s) formed as a result.
- Balanced equation:** Equation with equal atoms of each element on both sides.
- Oxidation:** Addition of oxygen or loss of electrons.
- Reduction:** Removal of oxygen or gain of electrons.
- Exothermic:** Releases heat to surroundings.
- Endothermic:** Absorbs heat from surroundings.
- Corrosion:** Slow destruction of metals by reaction with environment.
- Rancidity:** Oxidation of fats and oils causing bad smell and taste.
- Precipitate:** Insoluble solid formed during reaction.

3 Summary: Major Reaction Types

Reaction Type	General Form	One Balanced Example	Quick Observation / Colour
Combination	$A + B \rightarrow AB$	$2Mg + O_2 \rightarrow 2MgO$	White solid (MgO)
Decomposition	$AB \rightarrow A + B$	$CaCO_3 \rightarrow CaO + CO_2$	CO_2 gas; lime turns milky
Displacement (Single)	$A + BC \rightarrow AC + B$	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$	Blue $\rightarrow$ colourless; Cu reddish-brown
Double Displacement (Precipitation)	$AB + CD \rightarrow AD + CB$	$AgNO_3 + NaCl \rightarrow AgCl \downarrow + NaNO_3$	White ppt (AgCl)
Double Displacement (Gas Formation)	$AB + CD \rightarrow AD + CB$	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$	CO_2 gas; lime turns milky
Redox	Oxidation + Reduction	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$	Blue $\rightarrow$ pale green; Cu reddish-brown
Combustion	Fuel + $O_2 \rightarrow$ Oxides + Heat/Light	$2H_2 + O_2 \rightarrow 2H_2O$	Burns with 'pop' sound; water forms

4 12 Must-Remember Balanced Equations

1.	$2H_2 + O_2 \rightarrow 2H_2O$	Burns with 'pop' sound; water forms
2.	$2Mg + O_2 \rightarrow 2MgO$	White powder (MgO)
3.	$CaCO_3 \rightarrow CaO + CO_2$	CO_2 gas; lime turns milky
4.	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$	Blue $\rightarrow$ colourless; Cu reddish-brown
5.	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$	Blue $\rightarrow$ pale green; Cu reddish-brown
6.	$AgNO_3 + NaCl \rightarrow AgCl \downarrow + NaNO_3$	White precipitate (AgCl)
7.	$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 \downarrow + 2NaCl$	White precipitate (BaSO ₄)
8.	$Pb(NO_3)_2 + 2KI \rightarrow PbI_2 \downarrow + 2KNO_3$	Bright yellow precipitate (PbI ₂)
9.	$HCl + NaOH \rightarrow NaCl + H_2O$	Neutralisation; no colour change
10.	$2Cu + O_2 \rightarrow 2CuO$	Black solid (CuO)
11.	$CuO + H_2 \rightarrow Cu + H_2O$	Black $\rightarrow$ reddish-brown Cu; water forms
12.	$4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O$	Rust forms; reddish-brown hydrated oxide

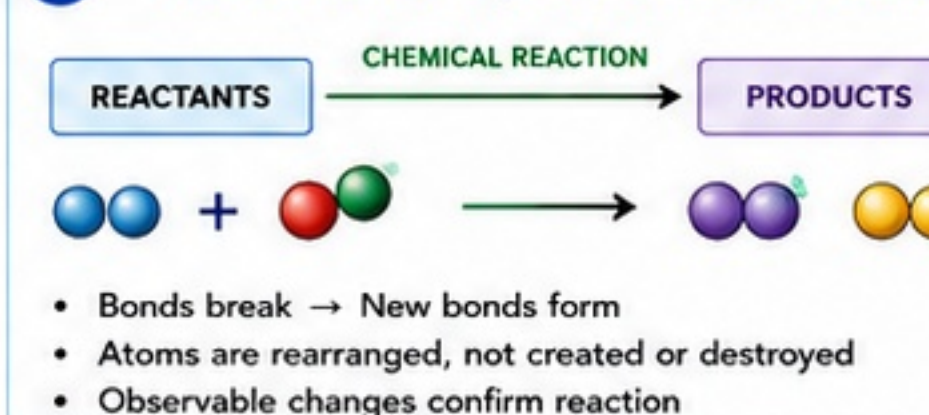
5 Common Mistakes Students Make

- Not balancing equations correctly.
- Writing states (s, l, g, aq) in the wrong places.
- Confusing reactants with products.
- Ignoring observations/colour changes in answers.
- Forgetting precipitate formation.
- Not using symbols and formulae properly.
- Writing ionic equations instead of balanced molecular equations when asked.

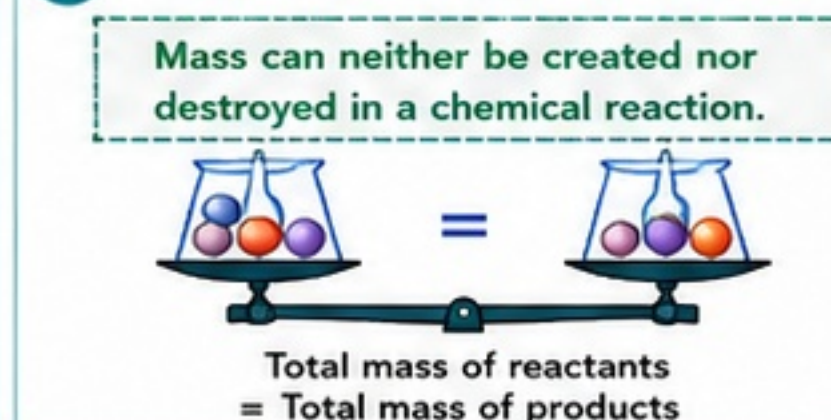
6 Exam-Winning Tips

- Always balance equations before writing.
- Mention states (s, l, g, aq) wherever possible.
- Write observations concisely (colour, gas, precipitate, heat, light).
- Use colour names exactly (e.g., pale green, bright yellow, reddish-brown).
- Learn common precipitates and gas tests.
- Practice writing equations from word statements and observations.

7 Quick Visual Recap: Reactants to Products



8 Law of Conservation of Mass



9 Formulas & Symbols Quick Reference

(s)	Solid	(l)	Liquid
(g)	Gas	(aq)	Aqueous
$\uparrow$	Gas evolved	$\downarrow$	Precipitate formed
Δ	Heat	h ν	Light
+	Plus / And	$\rightarrow$	Yields / Forms

Common Ions: Na^+ , K^+ , NH_4^+ , Ag^+ , Cu^{2+} , Fe^{2+} , Fe^{3+} , Zn^{2+} , Pb^{2+} , Ca^{2+} , SO_4^{2-} , NO_3^- , CO_3^{2-} , Cl^- , OH^- , I^- , S^{2-}

10 Revision Checklist

- ☐ I can define a chemical reaction.
- ☐ I can write and balance equations.
- ☐ I can identify type of reaction.
- ☐ I can predict observations.
- ☐ I can write equations from word statements.
- ☐ I know common precipitates and gases.
- ☐ I understand conservation of mass.
- ☐ I can explain redox in simple terms.

11 Self-Check Tracker

Rate Your Confidence (1 = Low, 2 = Medium, 3 = High)

Topic	1	2	3
Balancing Equations			
Types of Reactions			
Observations & Colours			
Precipitates & Gases			
Redox Concepts			
Real-Life Applications			
Overall Confidence			

Total Score (out of 21): _____ / 21

12 Challenge Yourself – Quick Recall

- Why must a chemical equation be balanced?
- What is the difference between reactant and product?
- Name the gas that turns lime water milky.
- What colour change is seen when iron reacts with $CuSO_4$ solution?
- Write the balanced equation for the reaction between $BaCl_2$ and Na_2SO_4 .
- Why does silver chloride form a white precipitate?
- Identify the type of reaction: $CaCO_3 \rightarrow CaO + CO_2$.
- What happens when copper(II) oxide is heated with hydrogen?
- Why do metals corrode? Give one example.
- State the law of conservation of mass.

★ CHEMISTRY IS NOT JUST A SUBJECT — IT'S THE REACTION THAT CREATES YOUR SUCCESS! ★